Emotional Shocks and Consumer Spending*

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Abstract

This study investigates the impact of emotional shocks on consumer spending behavior, using unexpected outcomes of professional football games as a natural experiment. By analyzing consumer scanner data, we examine changes in spending on food, non-food items, and alcohol following these emotional events. The dataset spans National Football League (NFL) regular seasons from 2004 to 2019, linking game outcomes to the purchase records of 83,332 households. Using Las Vegas pregame point spreads to identify emotional shocks, we find that unexpected losses by home teams lead to significant increases in household shopping trips and spending across all categories, while unexpected wins result in reduced spending, though with smaller magnitudes. Losses in closely contested games show positive but statistically insignificant effects. These emotional impacts are concentrated within three days post-game and amplified for high-stakes games. Our results are supported by various heterogeneity analyses and robustness checks. These findings provide insights into how emotional shocks from unexpected events can influence consumer spending behavior.

Keywords: emotional state, consumer behavior, consumption

JEL Codes: D12, D91, L83

*Disclaimer: Researcher(s)' own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

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"The thrill of victory...

and the agony of defeat...

the human drama of athletic competition"

— Jim McKay, ABC Wide World of Sports (1961)

1 Introduction

How do emotions influence consumers' shopping behaviors? Consumers' decisions are often shaped by the conditions under which they are made, and emotions can play a crucial role in this process (Williams et al., 2014; Iyer et al., 2020; Sharma et al., 2023). Anxiety during crises, for example, has been linked to stockpiling and changes in food purchases (Cai et al., 2024; Hobbs, 2020; Mahajan and Tomar, 2021), while happiness and contentment are associated with increased shopping, often encouraged by sensory strategies like scents and visual displays (Mattila and Wirtz, 2001; Mohan, Sivakumaran, and Sharma, 2013). Despite these insights, a significant gap remains in understanding the causal effects of emotional shocks—unexpected events that elicit intense emotional responses—on consumer spending and the differential impacts of positive versus negative emotions on shopping behaviors.

In this paper, we examine how emotional shocks influence consumer spending behavior, leveraging unexpected outcomes of professional football games as a natural experiment. We hypothesize that unexpected negative outcomes of football games trigger negative feelings in fans, while unexpected positive outcomes elicit positive feelings. Using detailed shopping data linked to National Football League (NFL) games, we analyze whether these emotional shocks affect consumers' shopping decisions and explore how the direction of the shock—positive or negative—differentially impacts spending behavior.

The NFL games, with their immense cultural significance to American fans, provide a unique context for examining how changes in emotional states influence subsequent purchasing behaviors. Several factors justify the focus on NFL games. First, NFL games are widely recognized as the most popular sporting events in the United States. Statistics show that the 2023 regular season attracted an average of 17.9 million viewers per game across all networks.[1] Therefore, the emotional reactions tied to NFL games reflect an intense, common, and recurring emotional experience for individuals. Second, NFL games have highly dedicated local fan bases compared to other sports events (World Sports Network, 2022a,b), capturing a significant share of the local TV audience. This strong fan attachment to local teams facilitates support identification. Third, the extended game season, structured competition, detailed statistics, and well-organized betting market enhance the ability to identify effects and explore mechanisms. Substantial variations in team performance across seasons also provide an ideal setting to investigate the effects of football events.[2] Last, the unexpected outcomes of football games have been used to measure emotional shocks in well-established studies, such as Card and Dahl (2011) and Eren and Mocan (2018).

We use the Las Vegas pregame point spread to represent fans' rational expectations of game outcomes and identify emotional shocks arising from deviations from this reference point. Specifically, unexpected outcomes that differ from the pregame point spread serve as either positive or negative emotional shocks for fans, allowing us to categorize events into upset losses (i.e., losses by a team when it was expected to win), close losses (i.e., losses by the team when the outcome was uncertain beforehand), and upset wins (i.e., wins by a team when it was expected to lose). To the extent that the pregame point spread efficiently predicts game outcomes, controlling for it enables us to interpret any differential impact between a win and a loss as the causal effect of the game outcome.

To investigate household purchase decisions, we utilize the NielsenIQ Homescan

[1] Source: NFL News, Jan 10, 2024. According to Nielsen's annual ranking of the 100 most-watched programs in 2023, NFL Games accounted for 93% of the most-watched TV programs and NFL games dominate weekly television ratings each fall.

[2] There are 14 different teams that have won the NFL championship in the past twenty years.

Panel, which provides comprehensive information on households and their buying behaviors. This dataset features a 19-year rotating panel of households, recording nearly all their purchases from major retailers and grocery stores. It includes detailed demographic data about households, as well as extensive information on products and purchases, such as prices, quantities, promotional activities, and a well-structured product hierarchy. Our study focuses on households with an exclusively local NFL team, resulting in a dataset comprising purchase records from 83,332 unique households during NFL regular seasons from 2004 to 2019.[3]

Using the Homescan dataset linked to NFL team records, we examine how emotional shocks from NFL games influence household spending, categorizing expenditures into three groups: food, non-food, and alcohol. Our analysis focuses on Sunday regular-season games, which account for approximately 86% of all such games during our study period. Rather than analyzing all Sunday regular-season games, we concentrate on high-stakes matchups—home games and division games—to better capture the effects of emotional shocks. The limited occurrence of back-to-back home and division games helps minimize confounding factors, such as closely timed shopping patterns, enabling a clearer interpretation of spending changes.[4] Table A1 presents NFL football teams, organized by divisions within the American Football Conference (AFC) and the National Football Conference (NFC). Conditioning on the Las Vegas pregame point spread, we regress the natural logarithm of spending on emotional shock variables, controlling for household characteristics and holiday dummies. We also include individual, week, and season-fixed effects in all specifications, which enables us to identify the causal effects of emotional shock exposure through random and exogenous variation in game outcomes.

The main findings of this study indicate that unexpected home team losses lead to

[3]The earliest available Nielsen Consumer Homescan data are from the 2004 panel, with the most recent data from the 2022 panel. We exclude data after the 2019 season to avoid potential confounding effects of the COVID-19 pandemic and subsequent price inflation.

[4]Results for all games and home games are reported in Appendix B. These findings are consistent with the main results, albeit with smaller magnitudes.

significant increases in both total spending and shopping trips, with spending rising across all categories and showing the greatest impact on food items. In contrast, unexpected wins lead to a reduction in spending. Losses in closely contested games show no significant effect on household spending. These changes in spending behavior are concentrated within a narrow window of 1–3 days.

We extend our analysis in several ways. First, we disaggregate food spending into perishable and non-perishable categories to examine potential substitution effects between food-at-home and food-away-from-home consumption. The results show that non-perishable food items experience larger changes in spending, suggesting that consumers may prioritize stocking durable goods following emotional shocks. Second, we analyze spending behavior by game type, focusing on the salience of each game. Our findings indicate that the effects of upset outcomes are particularly pronounced after high-stakes games, highlighting the importance of game intensity in driving household spending responses. Lastly, we investigate the heterogeneous effects of emotional shocks across demographic groups. We find that upset losses have a stronger impact on spending among households with older heads and higher disposable incomes. Meanwhile, upset wins show more pronounced effects on spending among households with non-white heads.

The findings remain consistent across several robustness checks. First, we employ alternative methods including different cutoffs for the point spread and using different parameterizations to define emotional shock. Second, we test alternative model specifications by varying fixed effects, control variables, and clustering levels to address potential concerns. Third, we analyze a restricted sample of counties with stronger attachment to the local team, further validating the results. Additionally, a placebo test confirms that household total spending is unaffected by home games scheduled for the following month.

Our findings connect with several strands of literature. The first strand delves into consumers' emotional shopping behaviors. A significant portion of this literature documents impulse buying, defined as a sudden and compelling urge to purchase goods immediately (Rook, 1987; Beatty and Ferrell, 1998). Impulse buying is shaped by internal factors such as personality traits and motives, as well as external stimuli like marketing strategies, with emotions playing a central mediating role (Redine et al., 2023; Iyer et al., 2020). While 80% of impulse purchases occur in physical grocery stores (Saleh, 2023), recent research has increasingly focused on online and social commerce platforms (Djafarova and Bowes, 2021; Huang, 2016; Sundström, Hjelm-Lidholm, and Radon, 2019; Yang et al., 2021), leaving impulsive behaviors in traditional grocery environments relatively underexplored (Lord, Putrevu, and Olson, 2023). Moreover, existing studies provide mixed evidence on whether impulse purchases are primarily driven by positive emotions, negative emotions, or both (Beatty and Ferrell, 1998; Mortimer, Bougoure, and Fazal-E-Hasan, 2015; Verhagen and van Dolen, 2011; Vohs and Faber, 2007), often relying on survey data with small sample sizes and self-reported emotional states, limiting causal insights.

Another strand of literature focuses on the impact of emotional states on consumption, particularly food choices under negative emotions. Stress and negative emotions have been shown to increase snacking and unhealthy eating (Gillebaart et al., 2022; Oliver and Wardle, 1999; Schultchen et al., 2019; Sproesser, Schupp, and Renner, 2014), with Cornil and Chandon (2013) linking fans' unhealthy eating habits following team losses to self-affirmation processes. More recently, broader shifts in consumption patterns driven by negative emotions have been documented during the COVID-19 pandemic (Gligorić et al., 2022; Cai et al., 2024).

We address gaps in these strands of literature by leveraging a comprehensive household scanner dataset that captures detailed in-store purchase records from a representative sample of consumers. This approach allows us to analyze how emotional shocks influence both the frequency and magnitude of spending across product categories, offering robust, real-world insights. By simultaneously measuring the effects of both negative and positive emotional shocks, we provide insights into their distinct impacts. Furthermore, the dataset's granular tracking of purchases enables us to directly link emotional changes to observable shopping behaviors, enhancing our understanding of how emotions drive consumer decisions in practical, high-stakes contexts.

Our study also intersects with research on the broader behavioral impacts of emotional shocks from unexpected sports outcomes. For instance, Card and Dahl (2011) and Cardazzi et al. (2022) show that heightened emotional responses to home football and basketball games significantly increase domestic violence rates in the host city. In contrast, Ivandić et al. (2024) find no significant effect of unexpected football results involving Manchester United and Manchester City on domestic abuse. Emotional shocks from sports outcomes also influence financial and academic outcomes. Edmans, García, and Norli (2007) report short-lived stock market declines after national soccer team losses, attributing these drops to changes in investor mood rather than economic factors. Similarly, Lindo, Swensen, and Waddell (2012) find that male students' GPAs decline during football seasons due to increased alcohol consumption and partying. Beyond these domains, sports-related emotional shocks have been shown to influence judicial sentencing (Eren and Mocan, 2018), asylum decisions (Chen and Spamann, 2014), and electoral outcomes (Healy, Malhotra, and Mo, 2010). Expanding this body of work, our findings document how emotional shocks from NFL games influence consumer spending behavior.

The remainder of this paper is organized as follows. Section 2 presents the data used in our analysis along with relevant summary statistics. Section 3 describes the econometric methodology, followed by a discussion of the core estimation results and several additional analyses and robustness checks. Finally, Section 5 provides a summary and concluding thoughts.

2 Data and Sample Construction

2.1 *NielsenIQ Homescan Data*

We use the NielsenIQ Homescan Panel covering 2004–2019 to measure household grocery purchases.[5] The dataset consists of a representative panel of households in the U.S., ranging from approximately 40,000 to 60,000 per year. These households use in-home scanners to record the Universal Product Codes (UPCs) of all consumer packaged goods they purchase from any retail outlet.[6] The Homescan data provides detailed information for each item purchased on each shopping trip, including transaction metadata such as purchase date, store/retailer information, and panelist identifier. Additionally, the Homescan Panel offers granular details on product and transaction details, including transaction price, amounts of quantities purchased, deal/sale usage, the total expenditure on the item, and detailed product identifiers.

NielsenIQ categorizes millions of UPCs into ten departments (i.e., the broadest category), six of which are food-related: dairy, deli, dry grocery, fresh produce, frozen food, and packaged meat. For our main analyses, we combine these six departments into a single "Food" category, representing approximately 65% of the total expenditure tracked by the Homescan Panel. The remaining departments are grouped as follows: general merchandise, health and beauty, and non-food grocery are combined into a "Non-food" category, while alcohol is classified as a separate, individual department.[7]

NielsenIQ recruits panelists using direct mail and Internet advertising and works to ensure the sample represents a broad demographic spectrum. To maintain data quality, NielsenIQ sets a purchasing threshold over a 12-month period to exclude households

[5]These data are available for academic research through a partnership with the Kilts Center at the University of Chicago Booth School of Business. Visit http://research.chicagobooth.edu/nielsen for more details.

[6]The data include purchases from supermarkets, convenience stores, mass merchandisers, club stores, drug stores, and other retail channels.

[7]Our analysis excludes "magnet" items without barcodes, as data for these products is unavailable before 2007 and only a small subset of households consistently report purchases of these items. To maintain consistency, we focus on packaged items.

that report only a minimal portion of their expenditures. Panelists are incentivized to remain engaged through regular prize drawings, sweepstakes, and a "gift points" system. The panel experiences an annual attrition rate of approximately 20%, with new households regularly recruited to replace those that leave.[8] On average, households remain in the panel for about 54 months.[9]

Homescan households annually report a wealth of demographic and socioeconomic characteristics, including age, education, marital status, race, household size, and income.[10] These attributes are categorized into different groups within the dataset. For our study, we reclassify some of these categories to serve as control variables.[11] For households with two heads, we use their average age and education levels.

While the Homescan Panel data offers extensive information, it has limitations relevant to our study. First, Homescan does not cover data on food purchased away from home, such as meals at fast food or full-service restaurants. This limitation prevents us from fully capturing changes in household spending, particularly in distinguishing between expenditures on food consumed at home and food consumed away from home. Research by Saksena et al. (2018) indicates that U.S. consumers obtain 34% of calories from food away from home, with a large portion consumed in restaurants where perishable items like meat and produce are common. To provide some insights, we categorize food into perishable items (e.g., dairy, deli products, fresh produce, and packaged meat) and non-perishable items (e.g., dry groceries and frozen foods). Since non-perishable

[8] NielsenIQ begins collecting data for new panelists on the first Sunday before the start of the new year, therefore the panel year does not perfectly align with the calendar year. For example, the 2004 panel includes data for 2003-12-28 through 2004-12-25.

[9] For a detailed description of the procedures for recruiting and retaining panelists, please refer to Muth, Siegel, and Zhen (2007) and the Consumer Panel Dataset Manual. This dataset features projection weights designed to ensure the sample is representative of the overall U.S. population. These weights enable researchers to extrapolate data to national, regional, and scantrack market levels. Bronnenberg et al. (2015) provide a summary supporting the representativeness of the panel of U.S. consumers.

[10] All demographics data is collected during the fall of the year prior to the panel year, except for household income data, which is for two years prior to the panel year.

[11] Age, education, and income information in NielsenIQ's consumer data are classified into 10, 7, 20 groups, respectively. Following Allcott et al. (2019), we assign the mean value of each group for age and income, and assign a representative number of years for each education group. For instance, an individual who has completed college is assigned 16 years of education.

items are less likely to have substitutes outside the home, we analyze how emotional shocks impact household spending on these two categories of food. Second, although Homescan records the date for each purchase, it does not specify the exact time of the transaction. This limitation affects our ability to determine whether a purchase was made before or after a game on the game day. To mitigate this issue, we assess whether the effects on consumer spending vary depending on the timing of the game within the day.

2.2 *Linking Homescan Panel to NFL Team Records*

We link the Homescan Panel to National Football League team records for local franchises. To identify local franchises, we follow the approach of Card and Dahl (2011). Specifically, we focus on states and cities with NFL teams. For states with only one NFL team, all panelists within that state are linked to that one NFL team. In states with multiple NFL teams located in different cities, panelists are assigned based on their city of residence to the corresponding local team. We exclude the New York Jets and New York Giants from our analysis due to their shared home city, as well as the Los Angeles Chargers and Los Angeles Rams due to their recent relocation.[12] Additionally, we exclude panelists who had ever moved to other cities without the same home team. Figure 1 illustrates the distribution of local NFL team franchises.

The NFL season begins with the regular season, which spans 17 weeks from early September to late December or early January. This is followed by the playoff season, which lasts about 4 weeks.[13] Due to the annual rotation of panelists and the inconsistent coverage of the final week of the regular season in the Homescan panel year, our analysis is restricted to the first 16 weeks of regular-season games, which account for 86% of

[12]The Chargers moved from San Diego to Los Angeles in the 2017 season, and the Rams moved from St. Louis to Los Angeles in the 2016 season.

[13]From the 2004 to 2019 seasons, there were a total of 7,893 NFL games, of which 95.88% (7,568) were regular-season games.

all regular-season games. To simplify our empirical design, we focus specifically on regular-season Sunday games. In addition, we restrict our analysis to home and division games due to their salience and the reduced likelihood of consecutive matchups, which minimizes confounding factors such as closely timed shopping patterns and stockpiling behavior. As Table 1 shows, there is a total of 987 games included in our sample. Having imposed these restrictions, our final sample includes 83,332 unique households matched to 30 NFL teams during the 2004–2019 period.[14]

Linking the Homescan panel data with the records of local NFL teams allows us to examine how consumer spending behaviors change during the week following a Sunday NFL game throughout the regular season. Table 2 offers a comprehensive summary of household expenditures and shopping trips and Table 3 reports households' demographic characteristics. We deflate expenditures and incomes to real 2012 dollars using the consumer price index for urban consumers for all items. Particularly, in Table 2, we present a descriptive summary of household expenditures across different categories on game day and compare them with spending patterns in the first two days afterward (Monday and Tuesday) and the subsequent week (Monday through Saturday). Additionally, we compare the average expenditure and the number of trips during game weeks with those during bye weeks when no game was played. On average, households in our sample spend approximately $104 each week.[15] The majority of this expenditure, over 60%, is on food, with only about 3.5% allocated to alcohol. This spending pattern is consistent across different time windows. During game weeks, households tend to spend more on alcohol and non-food items compared to during bye weeks, though their food expenditure is slightly reduced.

[14] Detailed observations for each NFL team over the years are presented in Table A1 in the appendix.

[15] Households in our sample spend an average of $66 per week on food. According to the U.S. Bureau of Labor Statistics (BLS), the average annual expenditure on food at home for all consumer units (with an average of 2.5 persons per unit) is $3,921. This figure is slightly higher than our calculation of $3,432, which can be partially attributed to our focus solely on packaged foods. Visit BLS Consumer Expenditures (Annual) News Release for more information.

2.3 *NFL Teams' Predicted and Actual Outcomes*

Sports spread betting involves predicting the outcome of a match based on a point spread, which indicates the expected score difference between two teams. In NFL games, betting on game outcomes is organized through Las Vegas bookmakers. For example, if Team A is favored by 3 points over Team B, Team A must win by more than 3 points for a bet on Team A to be successful. The closing value of the point spread captures the market's prediction of the game's outcome.

Card and Dahl (2011) and Eren and Mocan (2018) demonstrate that pregame point spreads are effective predictors of NFL game outcomes. Building on their findings, we gather data on pregame point spreads and final scores of all NFL games played during the 2004-2019 seasons[16] and present the relationship between the actual and predicted point spread in each game in Figure 2. A simple regression of the actual spreads on the predicted spreads yields a coefficient of 1.05 (standard error of 0.03) with a high R^2 value of 0.18. These findings suggest that the predicted point spreads are reliable predictors of NFL game outcomes.[17]

With the reliability of the point spreads established, we then categorize them into three distinct groups. Following the approach of Card and Dahl (2011) and Eren and Mocan (2018), we classify the games into three ex-ante categories, as shown in Figure 2: (i) predicted win if the point spread is -4 or less, (ii) predicted close if the point spread is between -4 and 4, and (iii) predicted loss if the point spread is 4 or more. We use different point spread cutoffs to assess the robustness of our results, which will be detailed in Section X.

Panel A and Panel B of Table 1 present a summary of the actual and predicted outcomes of the 987 regular-season Sunday home and division games included in our sam-

[16]The Spreadspoke company offers comprehensive data on NFL scores and betting information. The dataset can be accessed through Kaggle at the following link: NFL Scores and Betting Data.

[17]Our estimated effect closely matches the results reported by Card and Dahl (2011) and Eren and Mocan (2018).

ple. Of these games, 39.51% were predicted as wins, 15.50% as losses, and 44.98% as close matches. The teams lost approximately 25.38% of the games they were predicted to win by four or more points (i.e., upset losses) and won about 19.61% of the games they were predicted to lose by the same margin (i.e., upset wins). For games predicted to be close, approximately 50% ended in losses (i.e., close losses).

Table 1 also highlights important features of NFL games that are explored in subsequent analyses. In Panel C, we present the starting times of home and division games. A large share of these games (91%) started at or before 7 PM, indicating that only a small number of games were played at night. In Panel D, we consider the emotional salience of a game: whether the game had an unusually high number of sacks, turnovers, or penalty yards.[18] Approximately 55% of games featured a high number of sacks, turnovers, or penalty yards.

3 Empirical Methodology

To estimate the impact of emotional shocks generated by unexpected wins or losses from NFL games on consumer spending, we analyze regressions of household expenditures on emotional shock exposure variables. Building on existing research (e.g., Card and Dahl, 2011; Eren and Mocan, 2018; Ivandić et al., 2024), we specify the following equation:

$$\begin{aligned} Ln(Spending_{idks}) = \lambda_0 &+ \lambda_1 Upset\ Loss_{ks} + \lambda_2 Close\ Loss_{ks} + \lambda_3 Upset\ Win_{ks} \\ &+ \lambda_4 Predicted\ Win_{ks} + \lambda_5 Predicted\ Close_{ks} + \lambda_6 Predicted\ Loss_{ks} \\ &+ \mathbf{X}'_{iks}\beta + \mu_i + \gamma_k + \delta_s + \epsilon_{idks} \end{aligned} \tag{1}$$

[18] We classify games as salient if they feature frustrating occurrences such as five or more sacks, five or more turnovers, or 100 or more penalty yards.

where $Spending_{idks}$ is the amount of dollar spent on goods at home by household i on day d of week k in game season s. We focus on household spending on the day of the game, which is Sunday, as well as the two days following (Monday through Tuesday), and the rest of the week (Monday and Saturday). The variables *Predicted Win*, *Predicted Close*, and *Predicted Loss* are mutually exclusive indicators indicating the expected outcomes of a game based on the point spread in the betting market. These indicators represent whether the team is predicted to win, have a close game, or lose, respectively. The variables *Upset Loss*, *Close Loss*, and *Upset Win* are interactions between these predicted outcome dummies and the actual game results. Specifically, *Upset Loss* indicates that the team was predicted to win but ended up losing, while *Upset Win* indicates the opposite, where a loss was predicted, but the team won. *Close Loss* denotes a close game was predicted, but the team lost.[19] The variable $\mathbf{X}'_{iks}$ includes various observed household characteristics, including age, years of education, indicator of children, household size, marital status, race, income, and internet access. It also considers household spending prior to the games to control for potential stockpiling behavior and includes a holiday indicator to account for the changes in spending during national holidays. The term μ_i captures the household fixed effects; γ_k and δ_s denote week and season effects, respectively; and ϵ_{iks} is the error term in the model. Standard errors are clustered at the household level. Our results remain robust even when we cluster at the season $\times$ week level or use two-way clustering (i.e., at the household and season), as detailed in Section 4.

In this model, the weeks when no game was played (i.e., bye weeks) are the reference category. The key coefficients of interest are λ_1, λ_2, and λ_3, which capture the effects of an upset loss, a close loss, and an upset win on consumer spending, respectively. These estimates are similar to intention-to-treat (ITT) estimates because they include all

[19] In practice, we create the *Upset Loss*, *Close Loss*, and *Upset Win* as follows: $Upset\,Loss = \mathbb{1}(\text{Predicted Win}) \times \mathbb{1}(\text{Loss})$, $UpsetWin = \mathbb{1}(\text{Predicted Loss}) \times \mathbb{1}(\text{Win})$, $Upset\;\;Close = \mathbb{1}(\text{Predicted Close}) \times \mathbb{1}(\text{Loss})$.

households in the sample, regardless of whether they actually watched the NFL games. Positive values of these coefficients would imply that unexpected game outcomes lead to increased consumer spending, whereas negative values would suggest the opposite effect. Although the coefficients of predicted outcomes (i.e., λ_4, λ_5, and λ_6) are also noteworthy, they cannot be interpreted as causal because unobservable factors might simultaneously influence both the predicted outcomes and consumer spending decisions.

This framework relies on the key identifying assumption that, after accounting for the pregame point spread, the actual outcome of an NFL game is as good as random. If the Las Vegas spread effectively predicts NFL game outcomes, then controlling for the point spread in Equation (1) can yield unbiased estimates of the causal effect of emotional shocks resulting from game results.[20]

4 Results

In this section, we present our key findings on how emotional shocks impact household spending across various categories, both immediately and over time. We then examine the heterogeneity in the effects of emotional shocks on spending decisions. Next, we discuss several robustness checks using alternative specifications and sample selections to ensure the reliability of our results, followed by the results of falsification tests.

4.1 *Main Results*

Table 4 presents the main findings from estimating the model outlined in Equation (1), with total household spending as the dependent variable across three timeframes: Sun-

[20]Mathematically, for a binary random variable y representing the actual outcome of a game, with mean probability p, the expectation $E[y|p,Z] = E[y|p]$ holds true for any set of factors Z. In other words, when conditioned on the mean p, the actual outcome y is independent of Z. If the Las Vegas spread can effectively predict NFL game outcomes, we can express the mean probability p as a function of the point spread, $p(S)$. Assuming that $p(S)$ is invertible and independent of Z, we have $E[y|S,Z] = E[y|p,Z] = E[y|p]$. This indicates that the actual outcome y is independent of Z when conditioned on the point spread S (i.e., predicted outcomes).

day (game day), Monday to Tuesday, and Monday to Saturday, as shown in columns (1), (2), and (3), respectively.[21] The results in column (1) show that an upset loss by the home team during a home and division game leads to an immediate, statistically significant increase of 2.3% in household spending on Sunday. In contrast, a close loss is associated with a positive but statistically insignificant change in spending. An upset win, meanwhile, is estimated to reduce household spending by 1.3%, though this effect is not statistically significant.[22]

Behavioral responses to emotional shocks intensify during the first two days after game day (Monday through Tuesday). As shown in column (2), spending increases significantly by 5.2% following upset losses, more than doubling the effect observed on Sunday. Conversely, an upset win leads to a 4.1% reduction in spending, a significant change compared to Sunday, where the effect was insignificant. In addition, a close loss has no significant impact on household spending. Over the longer period of Monday to Saturday (column 3), the effects of emotional shocks largely diminish, with no significant changes in spending for either upset losses or wins. These findings suggest that the impact on spending is short-lived, with effects fading by the end of the week.

To further understand households' behavioral responses, we examine whether households change their grocery shopping frequency or merely alter their spending levels. To this end, we modify Equation (1) by replacing the dependent variable with the number of grocery store visits.[23] Given the incident-based nature of trip data, we employ a Poisson model for the estimation. The results, presented in columns (4), (5), and (6) of Table 4, are organized by observation windows. Our findings reveal that emotional shocks

[21] Estimates of the complete set of coefficients in Table 4 are presented in Table A3 the appendix.

[22] Since we do not have household-specific shopping times, we cannot determine whether Sunday purchases occurred before or after the game. To assess the potential impact of this limitation, we re-estimate the baseline specification separately for games starting before or at 1 PM and games starting before or at 7 PM. The results, presented in Table A5, show that the effects of unexpected game outcomes on household spending are similar regardless of whether games start early or late. This suggests that the timing of games on Sunday does not significantly influence total household spending.

[23] Shopping trips are defined by the frequency of visits made to purchase groceries. Each visit is counted as one trip, regardless of whether it involves stops at multiple stores.

significantly influence shopping trips. Specifically, an upset loss increases the number of grocery shopping trips by 1.3% on game day and by 3% over the following two days. In contrast, upset wins significantly reduce trips by 3.6% during the first two days following the game. We find no significant effects of close losses. Over the extended period of a full week after the game, the effects of emotional shocks dissipate. Neither upset losses nor wins have a statistically significant impact on shopping trips during this timeframe. These findings are consistent with the observed effects of emotional shocks on total household spending, suggesting that changes in grocery shopping frequency play a role in altering overall household expenditure.

We then analyze household spending across various categories, with the results presented in Figure 3. This figure shows the estimated effects of emotional shocks from NFL game outcomes on household spending across three categories: food, alcohol, and non-food items. The effects are measured over three time periods: Sunday (blue circles), Monday to Tuesday (green triangles), and Monday to Saturday (orange squares), with bars representing 95% confidence intervals. Emotional shocks, particularly upset losses, have the most pronounced impact on spending. Food spending increases significantly following an upset loss, with the largest effect occurring over Monday and Tuesday (4.33%) and a smaller yet significant increase on Sunday (1.70%). Non-food spending follows a similar pattern, rising by 2.16% over Monday and Tuesday and 1.70% on Sunday.[24] Alcohol spending also increases during these periods, albeit to a lesser extent, and its impact persists throughout the entire week.

In contrast, upset wins consistently lead to significant reductions in spending across all categories. Food spending drops by 3.88% over Monday and Tuesday and 2.67% on Sunday, while non-food spending declines by 3.33% during the first two days following the game day. Alcohol spending shows no meaningful changes after upset wins, suggest-

[24]While the NielsenIQ Homescan Data does not cover data on food purchased away from home, the observed impact on non-food purchases, which have fewer substitutes outside the home, provides further evidence that consumers' shopping decisions are influenced by emotional shocks.

ing that consumers neither reduce their alcohol intake under positive emotional shocks nor increase consumption to celebrate upset wins. Close losses, on the other hand, have negligible effects across all categories and timeframes, with small and statistically insignificant changes. These findings indicate that emotional shocks from unexpected game outcomes drive immediate adjustments in household spending, particularly for essential categories like food and non-food items, while the impacts diminish over the longer week following the game.

Overall, our main findings reveal three important insights. First, an unexpected loss by home teams in emotionally charged games leads to a significant increase in consumer spending, both on the game day and in the short period following the event. Conversely, losses anticipated to be close contests have a positive but limited impact on household spending, suggesting that less expected losses trigger stronger emotional responses than more anticipated ones. Second, we observe that an unexpected win has a negative impact on consumer spending. Unlike the effects of an upset loss, which are immediate, the impact of an upset win emerges after the game and is smaller in magnitude. Third, the changes in consumption spending can be partially attributed to changes in shopping trips. Finally, when examining different spending categories, we find that food expenditures are most sensitive to emotional shocks, while spending on alcohol and non-food items is also affected, albeit to a lesser extent.[25]

4.2 *Extension Analyses*

A. Spending across Different Food Categories

As noted in the Data and Sample Construction section, a key limitation of the Homescan Panel data is the absence of information on food purchased away from home. This prevents us from capturing shifts between spending on food at home and dining out.

[25]The main results for all games and home games are presented in Appendix Table B3 to Table B5. The findings reveal similar patterns to the main analysis, although the effects are smaller in magnitude.

For instance, a decrease in spending on food at home following an unexpected win might be due to an increase in dining out. To explore this potential substitution relationship, we categorize food into perishable and non-perishable items, as consumers are more likely to consume perishable food away from home.

Table 5 presents the relative effects of emotional outcomes on household food spending, with the first three columns focusing on perishable food and the last three columns on non-perishable food. We find that an upset loss leads to an increase in consumer spending on both perishable and non-perishable foods on the game day and for the two days following. However, the increase in spending on perishable foods is smaller compared to non-perishable foods. We also find that a close loss has a minimal positive impact on spending for both food categories. Conversely, an upset win results in a decrease in consumer spending on both perishable and non-perishable foods after the game, with a notably larger reduction in spending on non-perishable foods. The findings suggest that, to some extent, a substitution exists between spending on food at home and dining out. Nevertheless, emotional shocks still have a significant impact on perishable food expenditures at home.

B. Emotional Salience Games

Assuming the link between NFL game outcomes and household spending is driven by emotional shocks from unexpected results, it is reasonable to expect stronger effects for games with greater emotional intensity. To test this, we examine three potentially frustrating events: five or more sacks, five or more turnovers, or 100 or more penalty yards. At least one of these events occurs in approximately 50% of the games in our sample. Table 6 presents the effects of emotional shocks on household spending, comparing more emotionally salient games (columns 1–3) to less salient games (columns 4–6). For frustrating games defined in this manner, the estimated effect of an upset loss is 2% on Sunday, 9% over Monday and Tuesday, and 2.9% over the week following the game day, compared to 1.8% on Sunday, 4% over Monday and Tuesday, and no significant effect over the week for less frustrating games. These results indicate that households are more reactive to negative emotional shocks when games are more emotionally engaging, with the effects persisting throughout the week.

For close losses, there is a significant positive effect on spending for more salient games, with increases of 2.4% over Monday and Tuesday and 1.7% over the week. In contrast, close losses in less salient games do not significantly affect spending, suggesting that emotional intensity plays a critical role in driving consumer behavior following close contests. Upset wins show the opposite pattern, with spending decreasing by 7.3% on Sunday and 11.5% over Monday and Tuesday for more salient games. This reflects the dampening effect of an unexpected positive outcome in emotionally charged games. However, for less salient games, upset wins have no significant effect, indicating a muted emotional response when games are less engaging. These findings highlight the importance of game salience in shaping the magnitude and direction of household spending responses to emotional shocks.

C. Types of Households

We further explore households' emotional reactions to unexpected NFL game outcomes by household demographic characteristics: age (under 65 versus 65 and older), race (non-white versus white), and income level (below average vs. above average).

The first three columns of Table 7 display the results by the age of the household head. The findings for upset losses reveal mixed evidence across age groups. For younger households (under 65), an upset loss leads to significant increases in spending, with a 2.6% increase on Sunday and a larger 3.8% increase over Monday and Tuesday, suggesting heightened sensitivity to immediate negative outcomes. In contrast, older households (65 and older) exhibit a different pattern: spending increases are smaller and insignificant on Sunday but show a larger and statistically significant increase of

9.0% over the first two days following the game. This suggests that older households may delay their spending response to negative emotional shocks, potentially reflecting differences in spending habits or emotional processing between age groups. For upset wins, younger households reduce their spending by 3.6% over Monday and Tuesday, while older households show a slightly stronger reduction of 4.1% on Sunday. However, the *p*-value for the difference test exceeds 0.1, indicating that the differences between these groups are not statistically significant. Close losses have no significant impact on spending for either age group. Additionally, neither group exhibits lasting effects over the week in response to unexpected game outcomes.

Columns 4–6 of Table 7 present the results examining how households' emotional responses to unexpected game outcomes vary by race. Following an upset loss, non-white households increase spending by 2.1% on Sunday and 5.6% over Monday and Tuesday, compared to 3.3% on both Sunday and Monday and Tuesday for white households. However, the *p*-values from the difference tests indicate that these spending differences are not statistically significant. In response to an upset win, non-white households reduce their spending by 5.8% over Monday and Tuesday, a notably larger decrease than the smaller, insignificant reductions observed among white households. For close losses, both groups show insignificant changes in spending across all timeframes. Overall, the findings suggest that non-white households may be more sensitive to emotional shocks, particularly from unexpected positive outcomes.

Disposable income level also plays an important role in moderating spending responses to emotional shocks, though the results reported in the last three columns of Table 7 are mixed. Households with below-average income show larger spending changes on Sunday but smaller changes over Monday and Tuesday compared to their above-average-income counterparts. Specifically, below-average-income households increase spending by 2.7% on Sunday and 3.7% over Monday and Tuesday following an upset loss. In contrast, above-average-income households show a smaller, insignificant increase of 1.4% on Sunday, but a much larger increase of 7.4% over Monday and Tuesday. In the case of close losses, the spending responses are more nuanced. Below-average-income households exhibit smaller spending increases compared to above-average-income households immediately following the game, but their spending increases are larger over the subsequent two days. Following an upset win, below-average-income households significantly reduce spending by 4.8% during the first two days post-game, whereas above-average-income households show no significant change. These results indicate that households with lower income are generally more responsive to both positive and negative emotional shocks, particularly within the short-term window after the game.[26]

4.3 *Robustness of Spending Results*

We conducted several robustness checks to ensure the validity of our results. These included testing alternative parameterizations of Equation (1), exploring different model specifications, applying alternative sample selection and data cleaning methods, and analyzing a more restricted sample of households with stronger attachments to their local team.

A. Alternative parameterization

First, we keep the use of discrete parameterization and experiment with different cutoff values to define unexpected game outcomes. In our main specification, the cutoff values are -4 and 4. As a robustness check, we used alternative cutoff values of -3 and 3. The results, reported in Table A6, remain consistent with the main specification. Second, instead of using discrete parameterization as in Equation (1), we adopted an alternative approach following Card and Dahl (2011) and Eren and Mocan (2018). This method incorporates a cubic polynomial of the point spread and interacts it with an indicator

[26] The results of the extended analysis, covering both all games and home games, are presented in Appendix Table B6 and Table B7. The findings align with the main analysis, though the effects are generally smaller in magnitude.

for home team losses. Figure 4 visualizes the estimated interaction effect across a range of point spreads, along with the corresponding 95% pointwise confidence intervals. The results show that the effect of a loss on household total spending decreases as point spreads increase and is significantly different from zero when the pregame point spread is less than zero.

B. Alternative model specification

We further test the robustness of our results by replacing the season and week-of-season fixed effects with week-by-season fixed effects. The results, presented in Table A7, confirm the consistent impact of upset losses and close losses from NFL games on household spending, though we do not find significant effects associated with upset wins. In addition, we adjust the control for household spending prior to the game day. Instead of using spending on Saturday (the day before the game) as a control, we account for household spending over the entire week preceding the game. The results, reported in Table A8, remain consistent with our main findings. This indicates that precautionary inventory stocking before game day does not obscure the effects of emotional shocks on household spending. Furthermore, Table A9 reports the baseline estimates adjusted for alternative clustering. While our main analysis clusters standard errors at the household level, we replicate the results by clustering standard errors at both the household and game-season levels to assess whether clustering influences the estimated effects of emotional shocks. The results are nearly identical, suggesting that clustering does not present a significant issue in our analysis.

C. Dealing with potential data issues

Although NielsenIQ has a robust program for replacing panelists who fail to meet minimum reporting standards, we observe that approximately 10% of households report unusually zero total spending for over a month. To address this issue, we exclude these households and reestimate our main specification. The results, presented in Table A10, are consistent with our main findings. Additionally, in our main analysis, we exclude data from the final week of the regular season due to the annual rotation of panelists and inconsistent coverage in the Homescan panel during this period. As a robustness check, we incorporate household spending data from the final week of the regular season. For long-term panelists, we use their consecutive spending records, including the final week. For panelists with only one year of data, we estimate their spending in the final week based on their average day spending over the year. The results, reported in Table A11, remain consistent with the main findings, further supporting the robustness of our analysis.

D. A restricted sample with a stronger attachment

In our main analyses, we identify local franchises by focusing on states with an exclusive NFL team and home cities located in states with multiple NFL teams. We link all panelists in these states or cities to their respective local teams. However, recognizing that not all residents in states with an exclusive NFL team (e.g., Illinois) exhibit strong attachment to their state team, we restrict our sample to include areas with stronger local team loyalty. To define fandom at the county level, we use multiple criteria: the location of the home stadium, a fan map based on the geographic distribution of Twitter followers[27], and ticket sales data[28]. We include panelists residing in the home counties of NFL teams, as well as nearby counties with a high share of local team followers on Twitter and significant ticket sales for the team. The results, presented in Table A12, reveal a similar pattern of emotional shocks on household spending compared to the main results, but with larger magnitudes. This suggests that the restricted sample, which captures a stronger attachment to local teams, amplifies the effects of emotional shocks on

[27] Data source: Twitter, NFL fan map: Where are your team's followers? Accessed on December 1, 2024.
[28] Data source: Vividseats, NFL Fan Map: How America Roots for the NFL. Accessed on December 1, 2024.

spending behavior.

4.4 *Falsification Tests*

We performed a falsification test to investigate whether households' shopping decisions in a given week are influenced by the game results of the following month.[29] The results, presented in Table A13, replicate our benchmark regression from Table 4 (columns 1–3). All point estimates are small in magnitude and not statistically significant at the 5% level. These findings indicate that, as expected, NFL game results in the following month have no impact on household spending. The overall lack of statistical significance across the 18 coefficients reported in the table supports the robustness of our core results and suggests they are not spurious.[30]

5 Conclusion

In this paper, we leverage the unexpected outcomes of NFL games as a natural experiment to examine how positive and negative emotional shocks impact household spending in a real-world setting. Using the Las Vegas bookmakers' pregame point spread as a measure of fans' rational expectations about game outcomes, we analyze how unexpected results—such as upset losses, close losses, and upset wins—affect household consumption decisions.

Our findings show that emotional shocks from unexpected NFL game outcomes significantly affect household spending, though these effects are short-lived. Negative shocks, such as upset losses, lead to an immediate 2.3% increase in spending on game day (Sunday), which intensifies to 5.2% on Monday and Tuesday before fading over the week. In contrast, positive shocks, such as upset wins, result in a 4.1% reduction in

[29]The rationale for using game results from the following month is to rule out potential effects from stockpiling behaviors.

[30]We replicate the robustness checks for both all games and home games, with the results presented in Appendix Table B8 to Table B15, and find consistent results.

spending on Monday and Tuesday, with no significant effect observed on Sunday. Close losses do not have a measurable impact on spending. These results suggest that negative emotional shocks elicit stronger and more immediate behavioral responses than positive shocks. Furthermore, consumers appear to increase spending as a coping mechanism in response to negative emotions, while we find no evidence of spending increases driven by celebratory or contentment-related behavior. Additionally, these changes in spending are primarily driven by increases in shopping trips.

Extending the analysis, we find that food expenditures are more affected than alcohol and non-food items, high-stakes games amplify spending responses, and emotional shocks have heterogeneous effects across households, with stronger impacts for certain demographic groups. A variety of robustness analyses and a falsification test demonstrate the robustness of the results.

From a broader perspective, this study provides empirical evidence that short-term emotions significantly influence consumption decisions in real-world settings, highlighting their role in shaping economic behavior. Specifically, our work contributes to the limited empirical literature on the causal relationship between emotions and spending, particularly across diverse product categories.

Our findings have important implications for both policymakers and businesses. For policymakers, our results suggest that emotional shocks from unexpected sporting outcomes can trigger significant short-term changes in household consumption patterns, with potential implications for understanding household responses to other unexpected events. For businesses in the retail sector, our results reveal systematic patterns in consumer spending following unexpected game outcomes, particularly in food and non-food categories during the first few days after games. These patterns suggest opportunities for businesses to better understand and respond to emotion-driven changes in consumer behavior. However, the documented relationship between emotional shocks and spending behavior also raises important considerations about consumer vulnerability during periods of heightened emotions.

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Tables

Table 1: Summary Statistics of Sunday Regular Season NFL Games, 2004-2019

Home & Division Games	Number of Games	Fraction of Category or Subcategory (%)
Panel A. Game Outcomes		
Loss	450	45.59
Win	537	54.41
Panel B. Predicted and Actual Outcomes		
Predicted Win: Point Spread $\leq$ -4	390	39.51
Actual Win	291	74.62
Actual Loss (Upset Loss)	99	25.38
Predicted Close: -4 < Point Spread < 4	444	44.98
Actual Win	216	48.65
Actual Loss (Close Loss)	228	51.35
Predicted Loss: Point Spread $\geq$ 4	153	15.50
Actual Win (Upset Win)	30	19.61
Actual Loss	123	80.39
Panel C. Game Time		
Games Starting Before 1 PM	637	64.54
Games Starting Before 7 PM	902	91.39
Panel D. Salience Games		
Sacks $\geq$ 5, Turnovers $\geq$ 5, or Penalties > 100 yds	538	54.51

Notes: (1) This table presents summary statistics for NFL Sunday home and division football games during the first 16 weeks of each regular season from 2004 to 2019. (2) We exclude the New York Jets, New York Giants, Los Angeles Chargers, and Los Angeles Rams due to their shared home stadium locations. (3) Between 2004 and 2019, 5 regular-season Home & Division GamesNFL games played on Sundays ended in a tie. These ties were counted as a half-win and half-loss in league standings. We exclude these games in our analysis. (4) Table A2 in the appendix provides detailed regular-season win-loss records of each NFL team.

Table 2: Summary Statistics of Household Shopping Behaviors

	Game Week		Bye Week	
	Mean	Std. Dev.	Mean	Std. Dev.
Total Expenditure				
Sunday	17.00	47.81	17.40	47.95
Monday to Tuesday	19.91	43.32	19.90	43.07
Monday to Saturday	67.16	84.90	67.61	84.84
Food Expenditure				
Sunday	11.17	32.67	11.53	33.02
Monday to Tuesday	12.46	28.59	12.63	28.82
Monday to Saturday	42.08	54.07	42.89	54.75
Alcohol Expenditure				
Sunday	0.47	5.15	0.44	5.03
Monday to Tuesday	0.66	5.87	0.61	5.41
Monday to Saturday	2.50	11.92	2.32	11.55
Non-Food Expenditure				
Sunday	5.36	20.92	5.43	20.70
Monday to Tuesday	6.78	22.10	6.67	21.52
Monday to Saturday	22.58	43.64	22.41	42.70
Shopping Trips[a]				
Sunday	0.28	0.45	0.29	0.45
Monday to Tuesday	0.47	0.64	0.47	0.64
Monday to Saturday	1.46	1.33	1.47	1.33
Sample Size	787,724		388,416	

Notes: The summary statistics are unweighted for broader representativeness, as the final sample cannot be extrapolated to national, regional, or scantrack market levels where projection factors were applied in the HomeScan data. This table primarily presents the means and standard deviations of household expenditures and shopping trips following Sunday home and division games or bye weeks during regular seasons. We deflate expenditures to 2012 dollars using the consumer price index for urban consumers for all items. [a]Shopping trips are defined by the frequency of visits made to purchase groceries. Each visit is counted as one trip, regardless of whether it involves stops at multiple stores.

Table 3: Summary Statistics of Household Characteristics

Variable	Description	Mean	Std. Dev
Age	Household head's age	56.21	13.11
Education years	Household head's education years	14.46	1.97
Children	Indicates whether the household has children	0.23	0.42
Household size	The number of household members	2.35	1.29
Married[a]	Indicates whether the household status is married	0.61	0.49
White	Indicates whether the race of the household is white	0.82	0.38
Income[b]	The real income of the household ($1,000)	58.60	34.01
Internet access	Indicates whether the household's internet is active	0.86	0.35
Number of Households	The unique agents in our sample	83,332	

Notes: The summary statistics are unweighted for broader representativeness, as the final sample cannot be extrapolated to national, regional, or scantrack market levels where projection factors were applied in the HomeScan data. In this table, we provide descriptions and summary statistics for household socio-demographic characteristics. For households with two heads, we report the mean age and education observed for both heads. We deflate incomes to 2012 dollars using the consumer price index for urban consumers for all items. [a]For single household heads, 67.56% are female and 32.44% are male. [b]We use income data from two years before the panel year. Using current year income data would significantly reduce the number of observations due to sample attrition. Nonetheless, our results remain consistent when we use the current year income data.

Table 4: The Effects of Emotional Shocks on Household Total Spending and Shopping Trips

Dependent Var:	Ln (Total Spending)			Shopping Trips		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.023***	0.052***	0.007	0.013**	0.030***	0.000
	(0.008)	(0.008)	(0.008)	(0.006)	(0.005)	(0.003)
Close Loss	0.005	0.008	0.005	0.008	0.003	-0.000
	(0.007)	(0.007)	(0.008)	(0.005)	(0.004)	(0.003)
Upset Win	-0.013	-0.041**	-0.011	-0.011	-0.036***	-0.005
	(0.017)	(0.017)	(0.019)	(0.014)	(0.012)	(0.007)
Predicted Win	-0.010*	-0.021***	0.012**	-0.009**	-0.010***	0.002
	(0.005)	(0.006)	(0.006)	(0.004)	(0.004)	(0.002)
Predicted Close	0.005	-0.002	0.003	0.005	0.003	0.002
	(0.006)	(0.006)	(0.007)	(0.005)	(0.004)	(0.002)
Predicted Loss	0.026***	0.024***	0.025**	0.019***	0.014**	0.009**
	(0.009)	(0.009)	(0.010)	(0.007)	(0.006)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.129	0.152	0.206	-	-	-
Observations	1,176,140	1,176,140	1,176,140	1,080,076	1,102,474	1,160,389

Notes: The sample is restricted to expenditures/shopping trips following Sunday home and division games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education years, household size, and real income. In addition, controls include an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends) and the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) in columns (1)-(3) or shopping trips on the last Saturday in columns (4)-(6). *Bye Weeks* is the reference category. Given the incident-based nature of trip data, we instead use the Poisson model (*PPMLHDFE* command in STATA@18) for the number of trips reported by households. Some observations are excluded in columns (4)-(6) because they are either singletons or separated due to fixed effects. *** p<0.01, ** p<0.05, * p<0.1.

Table 5: Emotional Shocks and Household Shopping on Different Types of Food

Dependent Var: Ln (total spending on...)	Perishable Food			Non-perishable Food		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.010**	0.026***	0.003	0.018***	0.040***	0.010
	(0.005)	(0.005)	(0.006)	(0.006)	(0.007)	(0.008)
Close Loss	0.004	0.009*	-0.000	0.002	0.010*	0.001
	(0.004)	(0.005)	(0.006)	(0.006)	(0.006)	(0.007)
Upset Win	-0.021**	-0.022**	-0.005	-0.024*	-0.038***	-0.010
	(0.011)	(0.011)	(0.014)	(0.014)	(0.014)	(0.017)
Predicted Win	-0.002	-0.015***	0.010**	-0.008*	-0.017***	0.008
	(0.004)	(0.004)	(0.005)	(0.004)	(0.005)	(0.005)
Predicted Close	0.004	-0.006	0.002	0.003	-0.009*	-0.000
	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)	(0.006)
Predicted Loss	0.016***	0.009	0.010	0.021***	0.021***	0.024***
	(0.006)	(0.006)	(0.007)	(0.007)	(0.008)	(0.009)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.126	0.129	0.206	0.125	0.132	0.192
Observations	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. Perishable food includes dairy, deli, fresh produce, and packaged meat departments. Non-perishable food includes dry grocery and frozen foods departments. These categorizations are based on product hierarchy structured by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household marital status, childbearing, race, and internet access, as well as household age, education, household size, and real income. Additional controls include the logarithm of spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** p<0.01, ** p<0.05, * p<0.1.

Table 6: Shocks from Emotionally Salient Games

Dependent Var: Ln(total spending on...)	More Salient Games			Less Salient Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.020*	0.092***	0.029**	0.023**	0.028**	-0.027**
	(0.011)	(0.011)	(0.012)	(0.012)	(0.012)	(0.013)
Close Loss	0.005	0.024**	0.017*	0.001	-0.011	-0.011
	(0.009)	(0.010)	(0.010)	(0.010)	(0.011)	(0.012)
Upset Win	-0.073***	-0.115***	-0.016	0.037	-0.013	-0.000
	(0.025)	(0.06)	(0.028)	(0.024)	(0.025)	(0.028)
Predicted Win	-0.005	-0.047***	-0.003	-0.009	-0.005	0.029***
	(0.007)	(0.008)	(0.008)	(0.007)	(0.008)	(0.008)
Predicted Close	0.007	-0.028***	-0.013	0.013	0.020**	0.018**
	(0.008)	(0.008)	(0.009)	(0.008)	(0.009)	(0.009)
Predicted Loss	0.041***	0.078***	0.059***	0.024**	-0.026**	-0.006
	(0.012)	(0.013)	(0.014)	(0.012)	(0.013)	(0.014)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.130	0.153	0.207	0.129	0.151	0.202
Observations	801,172	801,172	801,172	743,848	743,848	743,848

Notes: This table examines the impact of three potentially frustrating events: five or more sacks, five or more turnovers, or 100 or more penalty yards. The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** p<0.01, ** p<0.05, * p<0.1.

Table 7: Emotional Shocks and Household Total Spending by Age, Race, and Income Level

Dependent Var:	Age			Race			Income Level		
Ln (total spending ...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Panel A	<65			Non-white			Below Average		
(a) Upset Loss	0.026***	0.038***	0.006	0.021**	0.056***	0.008	0.027***	0.037***	0.002
	(0.009)	(0.009)	(0.010)	(0.008)	(0.009)	(0.009)	(0.009)	(0.010)	(0.010)
(b) Close Loss	0.004	0.005	0.001	0.005	0.012	0.012	-0.008	0.022**	0.009
	(0.008)	(0.008)	(0.009)	(0.007)	(0.008)	(0.008)	(0.009)	(0.009)	(0.010)
(c) Upset Win	-0.000	-0.036*	0.002	-0.022	-0.057***	-0.022	-0.011	-0.048**	-0.019
	(0.020)	(0.020)	(0.023)	(0.019)	(0.019)	(0.021)	(0.020)	(0.022)	(0.023)
Predicted Win	-0.014**	-0.014**	0.007	-0.010*	-0.022***	0.009	-0.019***	-0.010	0.021***
	(0.007)	(0.007)	(0.007)	(0.006)	(0.006)	(0.007)	(0.007)	(0.007)	(0.008)
Predicted Close	0.005	0.002	0.002	0.003	-0.009	-0.007	0.016**	-0.008	0.006
	(0.007)	(0.007)	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)	(0.008)	(0.008)
Predicted Loss	0.021**	0.027***	0.020*	0.027***	0.027***	0.027**	0.025**	0.022*	0.034***
	(0.010)	(0.010)	(0.012)	(0.010)	(0.010)	(0.011)	(0.011)	(0.011)	(0.012)
Observations	864,445	864,445	864,445	967,026	967,026	967,026	679,514	679,514	679,514
Panel B	65+			White			Above Average		
(a') Upset Loss	0.009	0.090***	0.007	0.033*	0.033*	0.000	0.014	0.074***	0.010
	(0.013)	(0.015)	(0.015)	(0.018)	(0.018)	(0.020)	(0.013)	(0.013)	(0.014)
(b') Close Loss	0.010	0.007	0.016	0.007	-0.011	-0.018	0.020*	-0.013	-0.004
	(0.012)	(0.014)	(0.014)	(0.015)	(0.016)	(0.017)	(0.011)	(0.011)	(0.012)
(c') Upset Win	-0.041	-0.041	-0.037	0.017	0.011	0.019	-0.019	-0.024	0.005
	(0.029)	(0.034)	(0.035)	(0.036)	(0.038)	(0.043)	(0.029)	(0.029)	(0.032)
Predicted Win	0.000	-0.038***	0.026**	-0.014	-0.019	0.024	0.003	-0.033***	0.003
	(0.009)	(0.011)	(0.011)	(0.013)	(0.013)	(0.015)	(0.009)	(0.009)	(0.010)
Predicted Close	0.002	-0.007	0.007	0.010	0.029**	0.044***	-0.009	0.008	0.003
	(0.010)	(0.012)	(0.012)	(0.014)	(0.014)	(0.015)	(0.010)	(0.010)	(0.010)
Predicted Loss	0.035**	0.018	0.043**	0.018	0.012	0.019	0.029**	0.026*	0.010
	(0.015)	(0.017)	(0.018)	(0.020)	(0.020)	(0.023)	(0.014)	(0.014)	(0.016)
Observation	311,660	311,660	311,660	209,102	209,102	209,102	496,444	496,444	496,444
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
p-value for row (a) = row (a')	0.181	0.038	0.956	0.330	0.864	0.761	0.504	0.008	0.261
p-value for row (b) = row (b')	0.665	0.408	0.344	0.562	0.372	0.224	0.078	0.029	0.485
p-value for row (c) = row (c')	0.526	0.912	0.348	0.219	0.006	0.334	0.345	0.766	0.529

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Figures

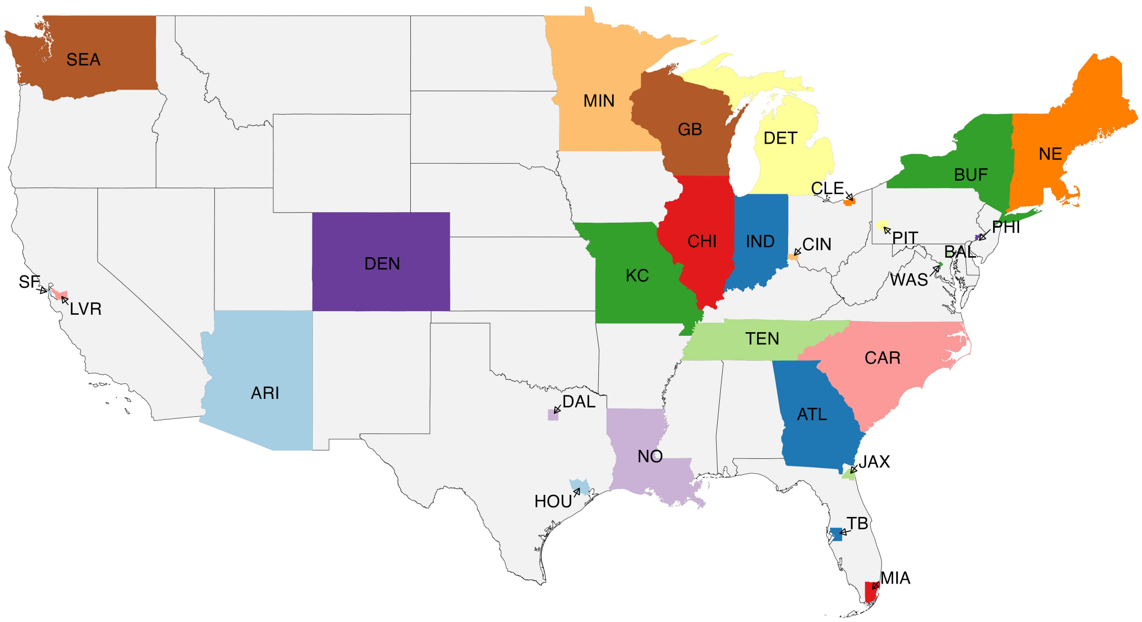

Figure 1: NFL Teams and Fan Map

Notes: This map illustrates the NFL teams included in the study sample and the geographic distribution of local franchises. For states with a single NFL team, all panelists within the state are associated with that team. In states with multiple NFL teams, panelists are assigned to the team corresponding to their city of residence.

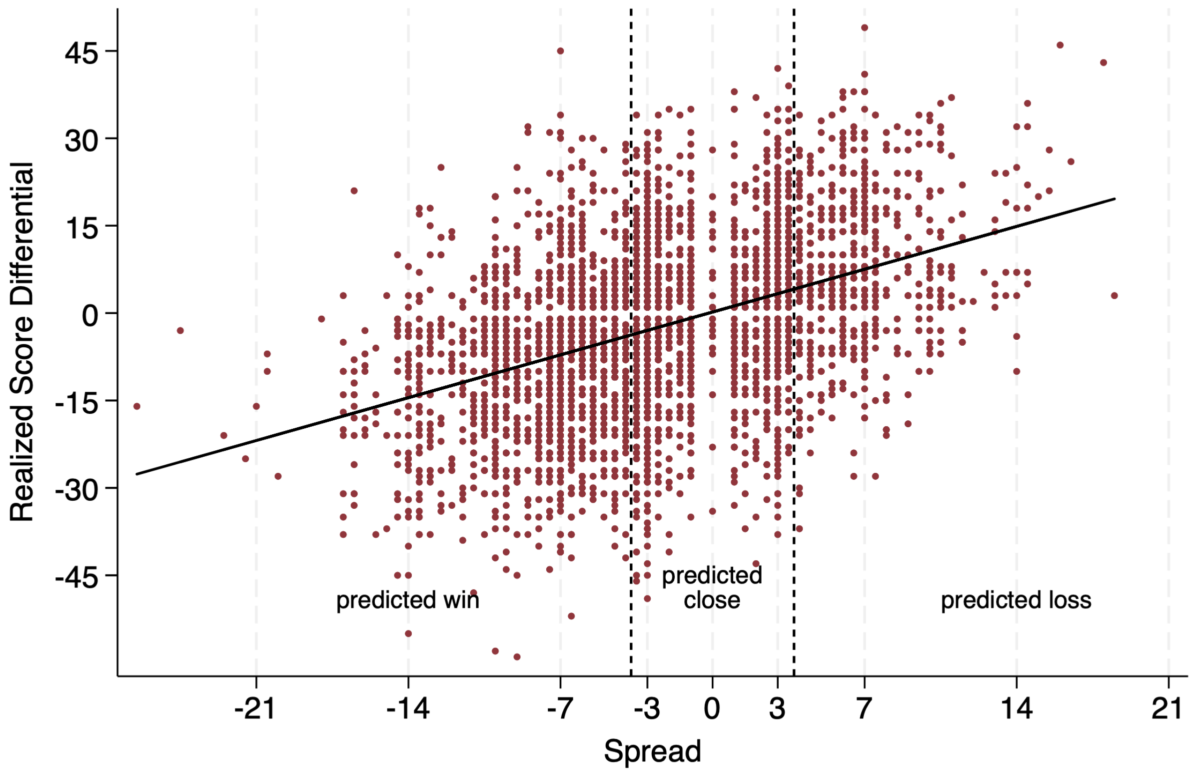

Figure 2: Final Score Differential versus the Pregame Point Spread

Notes: This figure illustrates the relationship between the actual and predicted point spreads for each NFL game during the 2004-2019 seasons. The regression line plotted has a slope of 1.05 (standard error = 0.03).

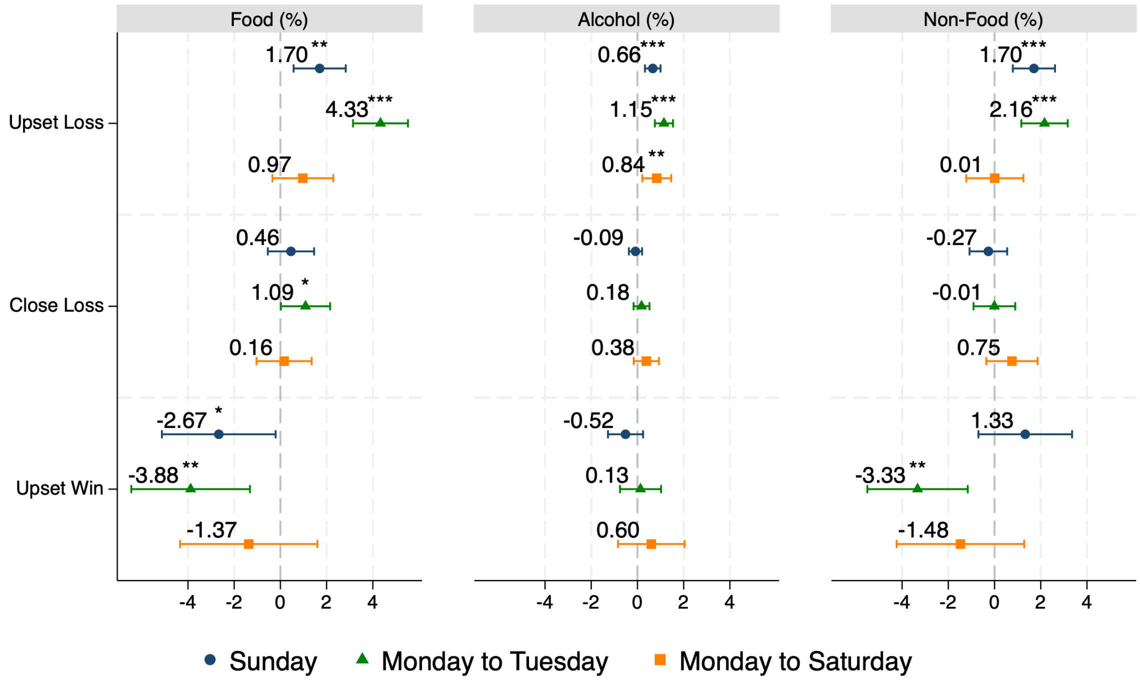

Figure 3: Emotional Shocks and Household Spendings on Different Categories

Notes: This figure illustrates the estimated coefficients for the emotional shocks on household total spending. The bars represent 95 percent confidence intervals, and standard errors are clustered at the household level. *** p<0.01, ** p<0.05, * p<0.1. Table A5 provide further information.

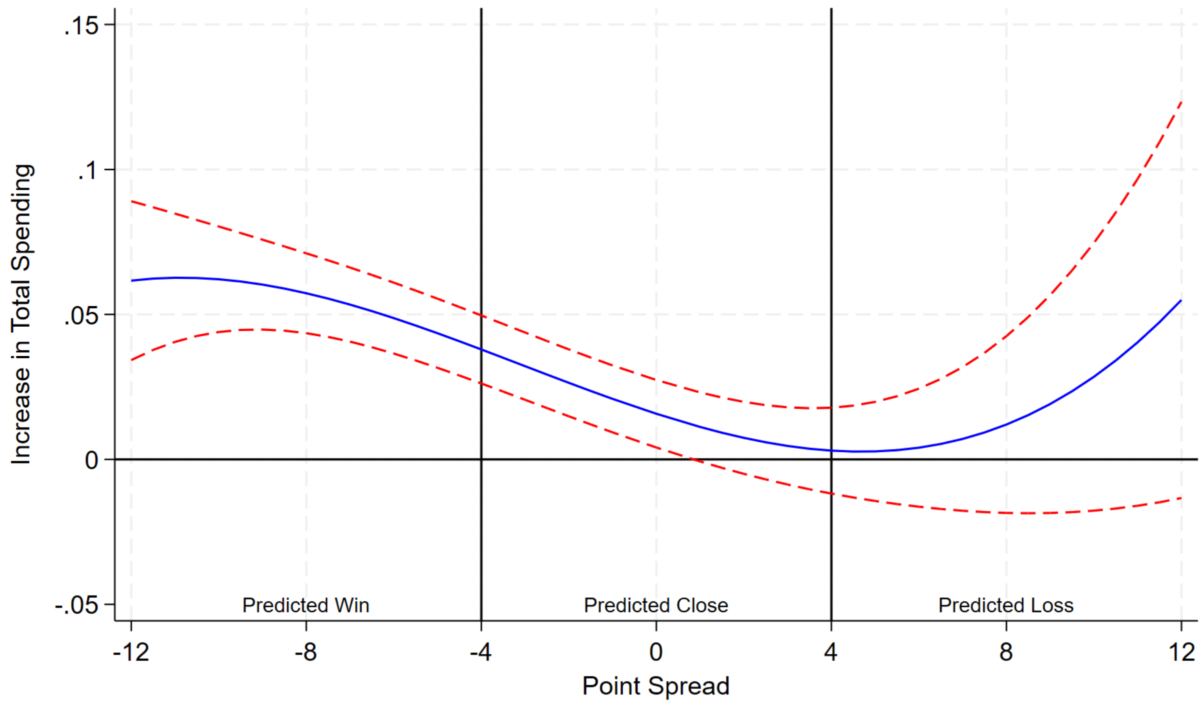

Figure 4: Increase in Total Spending for a Loss versus a Win as a Function of the Pregame Point Spread

Notes: The estimates are derived from a specification with a cubic-order polynomial in the point spread and an indicator variable for team loss. The dependent variable is the household total spending during Monday and Tuesday following a Sunday home and division game, or bye week during the regular NFL football season from 2004 to 2019. The dashed lines represent pointwise 95 percent confidence intervals.

Appendix A Appendix for Main Text

Table A1: NFL Football Teams

Conference	Division	Team	Team ID
AFC	EAST	Buffalo Bills	BUF
		Miami Dolphins	MIA
		New England Patriots	NE
		New York Jets	NYJ
	NORTH	Baltimore Ravens	BAL
		Cincinnati Bengals	CIN
		Cleveland Browns	CLE
		Pittsburgh Steelers	PIT
	SOUTH	Houston Texans	HOU
		Indianapolis Colts	IND
		Jacksonville Jaguars	JAX
		Tennessee Titans	TEN
	WEST	Denver Broncos	DEN
		Kansas City Chiefs	KC
		Los Angeles Chargers	LAC
		Oakland Raiders	LVR
NFC	EAST	Dallas Cowboys	DAL
		New York Giants	NYG
		Philadelphia Eagles	PHI
		Washington Redskins	WAS
	NORTH	Chicago Bears	CHI
		Detroit Lions	DET
		Green Bay Packers	GB
		Minnesota Vikings	MIN
	SOUTH	Atlanta Falcons	ATL
		Carolina Panthers	CAR
		New Orleans Saints	NO
		Tampa Bay Buccaneers	TB
	WEST	Arizona Cardinals	ARI
		Los Angeles Rams	LAR
		San Francisco 49ers	SF
		Seattle Seahawks	SEA

Notes: The United States has two major professional football conferences: the American Football Conference (AFC) and the National Football Conference (NFC). Each conference is divided into four divisions, with each division comprising four teams.

Table A2: NFL Football Teams Matched to Households in Sample (2004 - 2019)

Team	**2004**	**2005**	**2006**	**2007**	**2008**	**2009**	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**
Arizona Cardinals	1,430	1,371	1,608	1,768	1,671	1,615	1,694	1,642	1,605	1,618	1,546	1,542	1,570	1,531	1,512	1,486
Atlanta Falcons	1,349	1,342	1,764	1,950	1,925	1,921	2,035	1,981	1,941	2,049	2,089	2,127	2,116	2,064	2,129	2,203
Baltimore Ravens	76	77	107	109	109	115	116	111	116	119	115	114	114	114	111	117
Buffalo Bills	326	305	341	356	347	333	318	287	266	252	243	231	225	207	207	203
Carolina Panthers	2,020	1,968	3,185	3,521	3,483	3,552	3,754	3,548	3,492	3,560	3,530	3,535	3,531	3,619	3,686	3,594
Chicago Bears	2,040	2,005	2,819	3,093	3,152	3,389	3,416	3,174	3,154	3,172	3,134	3,135	3,224	3,069	2,976	3,007
Cincinnati Bengals	69	65	200	222	208	212	227	206	204	207	215	218	215	210	221	201
Cleveland Browns	77	71	147	158	174	176	171	163	170	176	179	175	164	165	169	172
Dallas Cowboys	151	151	165	172	166	176	188	169	169	166	175	184	180	173	167	168
Denver Broncos	1,310	1,275	1,557	1,634	1,495	1,441	1,500	1,390	1,348	1,361	1,350	1,348	1,405	1,377	1,341	1,302
Detroit Lions	1,448	1,474	2,436	2,726	2,623	2,722	2,761	2,609	2,611	2,610	2,580	2,651	2,698	2,643	2,600	2,570
Green Bay Packers	621	631	1,547	1,751	1,686	1,643	1,672	1,608	1,612	1,679	1,704	1,665	1,667	1,649	1,613	1,593
Houston Texans	542	510	570	578	529	525	539	519	513	535	547	550	554	528	523	502
Indianapolis Colts	591	588	1,604	1,805	1,723	1,738	1,721	1,667	1,640	1,670	1,685	1,696	1,715	1,705	1,692	1,648
Jacksonville Jaguars	60	57	163	177	165	160	161	150	152	171	174	191	182	175	178	169
Kansas City Chiefs	46	49	100	124	132	132	157	161	165	171	165	169	165	170	168	185
Miami Dolphins	241	250	252	241	242	252	243	242	248	249	236	241	239	223	222	225
Minnesota Vikings	1,427	1,409	1,749	1,857	1,798	1,708	1,769	1,654	1,589	1,611	1,544	1,557	1,552	1,480	1,458	1,440
New England Patriots	2,054	1,938	2,901	3,176	3,160	3,283	3,340	3,202	3,257	3,340	3,355	3,408	3,476	3,376	3,310	3,298
New Orleans Saints	696	588	851	877	826	849	841	802	815	849	860	878	918	924	949	933
Oakland Raiders	78	75	78	76	71	76	80	71	70	71	64	67	62	57	54	54
Philadelphia Eagles	202	186	213	233	223	239	248	245	248	258	248	269	272	250	251	268
Pittsburgh Steelers	80	77	216	242	226	222	225	222	220	220	221	207	217	203	206	207
San Diego Chargers	77	67	236	254	229	237	255	232	224	235	228	238	206	-	-	-
San Francisco 49ers	161	149	149	145	134	134	137	137	130	133	138	136	140	134	128	126
Seattle Seahawks	1,520	1,467	1,822	1,913	1,810	1,747	1,762	1,720	1,693	1,684	1,698	1,693	1,675	1,590	1,532	1,462
St. Louis Rams	524	498	513	515	508	463	465	419	399	376	340	302	-	-	-	-
Tampa Bay Buccaneers	134	133	154	175	180	165	169	163	162	153	148	169	155	148	153	158
Tennessee Titans	1,012	1,068	1,584	1,683	1,564	1,573	1,618	1,541	1,531	1,558	1,586	1,611	1,623	1,629	1,606	1,574
Washington Redskins	86	81	93	91	86	90	86	80	79	87	91	89	91	83	82	89
Total	**20,448**	**19,925**	**29,124**	**31,622**	**30,645**	**30,888**	**31,668**	**30,115**	**29,823**	**30,340**	**30,188**	**30,396**	**30,351**	**29,496**	**29,244**	**28,954**

Table A3: The Effects of Emotional Shocks on Household Total Spending and Shopping Trips (Full Table)

Dependent Var:	Ln (Total Spending)			Shopping Trips		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
(a)Upset Loss	0.023***	0.052***	0.007	0.013**	0.030***	0.000
	(0.008)	(0.008)	(0.008)	(0.006)	(0.005)	(0.003)
Close Loss	0.005	0.008	0.005	0.008	0.003	-0.000
	(0.007)	(0.007)	(0.008)	(0.005)	(0.004)	(0.003)
(b)Upset Win	-0.013	-0.041**	-0.011	-0.011	-0.036***	-0.005
	(0.017)	(0.017)	(0.019)	(0.014)	(0.012)	(0.007)
Predicted Win	-0.010*	-0.021***	0.012**	-0.009**	-0.010***	0.002
	(0.005)	(0.006)	(0.006)	(0.004)	(0.004)	(0.002)
Predicted Close	0.005	-0.002	0.003	0.005	0.003	0.002
	(0.006)	(0.006)	(0.007)	(0.005)	(0.004)	(0.002)
Predicted Loss	0.026***	0.024***	0.025**	0.019***	0.014**	0.009**
	(0.009)	(0.009)	(0.010)	(0.007)	(0.006)	(0.004)
Ln(Total Spending on Last Sat)/ Trips on Last Sat	-0.078***	-0.032***	0.018***	-0.169***	-0.028***	0.042***
	(0.001)	(0.001)	(0.001)	(0.004)	(0.003)	(0.002)
Age	-0.000	0.001	0.003**	-0.001	0.000	0.001*
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
Education years	0.000	0.005	0.005	-0.001	0.002	0.000
	(0.004)	(0.004)	(0.005)	(0.003)	(0.003)	(0.002)
Children	0.035**	0.093***	0.084***	0.018*	0.061***	0.042***
	(0.014)	(0.015)	(0.017)	(0.010)	(0.009)	(0.006)
Household Size	0.020***	0.020***	0.025***	0.011***	0.009**	0.007***
	(0.005)	(0.006)	(0.006)	(0.004)	(0.003)	(0.002)
Married	0.047***	0.135***	0.185***	0.035***	0.080***	0.080***
	(0.015)	(0.016)	(0.018)	(0.012)	(0.010)	(0.007)
White	0.043	-0.028	-0.031	0.029	-0.016	-0.010
	(0.028)	(0.029)	(0.032)	(0.021)	(0.019)	(0.013)
Ln(income)	0.038***	-0.028***	-0.022**	0.022***	-0.024***	-0.014***
	(0.007)	(0.008)	(0.009)	(0.006)	(0.005)	(0.004)
Internet access	-0.011	-0.013	-0.025*	-0.015	-0.008	-0.007
	(0.012)	(0.013)	(0.014)	(0.009)	(0.008)	(0.006)
Holiday week	-0.019***	-0.044***	0.013*	-0.012**	-0.027***	0.007***
	(0.006)	(0.007)	(0.007)	(0.005)	(0.004)	(0.003)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.129	0.152	0.206	-	-	-
Observations	1,176,140	1,176,140	1,176,140	1,080,076	1,102,474	1,160,389

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. *Bye Weeks* is the reference category. Some observations are excluded in columns (4)-(6) because they are either singletons or separated due to fixed effects. *** p<0.01, ** p<0.05, * p<0.1.

Table A4: Emotional Shocks and Household Spendings on Different Categories

Dependent Var:	Food Spending			Alcohol Spending			Non-Food Spending		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.017**	0.043***	0.010	0.007***	0.011***	0.008**	0.017***	0.022***	0.000
	(0.007)	(0.007)	(0.008)	(0.002)	(0.002)	(0.004)	(0.006)	(0.006)	(0.008)
Close Loss	0.005	0.011*	0.002	-0.001	0.002	0.004	-0.003	-0.000	0.008
	(0.006)	(0.006)	(0.007)	(0.002)	(0.002)	(0.003)	(0.005)	(0.005)	(0.007)
Upset Win	-0.027*	-0.039**	-0.014	-0.005	0.001	0.006	0.013	-0.033**	-0.015
	(0.015)	(0.016)	(0.018)	(0.005)	(0.005)	(0.009)	(0.012)	(0.013)	(0.017)
Predicted Win	-0.007	-0.020***	0.011*	-0.001	-0.007***	-0.009***	-0.009**	-0.011**	0.007
	(0.005)	(0.005)	(0.006)	(0.001)	(0.002)	(0.003)	(0.004)	(0.004)	(0.005)
Predicted Close	0.005	-0.008	0.003	0.001	-0.003*	-0.001	0.002	0.000	-0.005
	(0.005)	(0.006)	(0.006)	(0.002)	(0.002)	(0.003)	(0.004)	(0.005)	(0.006)
Predicted Loss	0.023***	0.021***	0.026***	0.005**	0.003	0.012***	0.008	0.015**	0.015*
	(0.008)	(0.008)	(0.009)	(0.002)	(0.003)	(0.004)	(0.006)	(0.007)	(0.009)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.129	0.143	0.200	0.088	0.129	0.277	0.096	0.114	0.195
Observations	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. Food includes dry grocery, frozen foods, dairy, deli, fresh produce, and packaged meat departments. Non-food includes health & beauty care, non-food grocery, and general merchandise departments. These categorizations are from the consumer panel data collected by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table A5: Timing of Shocks and Spending on Sunday

Dependent Var:	(1)	(2)
Ln (total spending on Sunday)	Games before 1pm	Games before 7pm
Upset Loss	0.025***	0.028***
	(0.009)	(0.008)
Close Loss	0.009	0.003
	(0.008)	(0.007)
Upset Win	-0.012	-0.016
	(0.019)	(0.017)
Predicted Win	-0.011*	-0.010*
	(0.006)	(0.006)
Predicted Close	0.004	0.005
	(0.007)	(0.006)
Predicted Loss	0.028***	0.022**
	(0.010)	(0.009)
Controls	Yes	Yes
Household FE	Yes	Yes
Season & Week of season FE	Yes	Yes
Adj.R^2	0.130	0.129
Observations	931,140	1,126,228

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household marital status, childbearing, race, and internet access, as well as household age, education, household size, and real income. Additional controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** p<0.01, ** p<0.05, * p<0.1.

Table A6: Emotional Shocks and Household Total Spending - Different Cutoffs

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.023***	0.049***	0.013*
	(0.007)	(0.007)	(0.007)
Close Loss	0.010	0.010	-0.010
	(0.011)	(0.011)	(0.012)
Upset Win	0.011	-0.017	-0.012
	(0.010)	(0.011)	(0.012)
Predicted Win	-0.008	-0.016***	0.009*
	(0.005)	(0.005)	(0.006)
Predicted Close	-0.004	-0.008	0.006
	(0.008)	(0.008)	(0.009)
Predicted Loss	0.020***	0.006	0.016**
	(0.007)	(0.007)	(0.008)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj.R^2	0.129	0.152	0.206
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -3 or less, *Predicted Close* indicates a point spread between -3 and 3 (exclusive), and *Predicted Loss* indicates a spread of +3 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A7: Emotional Shocks and Household Total Spending - Different Fixed Effects

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.025***	0.022**	0.013
	(0.009)	(0.010)	(0.010)
Close Loss	0.009	0.013	0.013
	(0.008)	(0.008)	(0.009)
Upset Win	-0.033	-0.011	0.004
	(0.020)	(0.021)	(0.023)
Predicted Win	-0.005	-0.001	0.015**
	(0.006)	(0.007)	(0.007)
Predicted Close	-0.000	-0.001	-0.004
	(0.007)	(0.007)	(0.008)
Predicted Loss	0.025**	0.021*	0.022*
	(0.011)	(0.011)	(0.012)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season - week FE	Yes	Yes	Yes
Adj.R^2	0.130	0.154	0.206
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table A8: Emotional Shocks and Household Total Spending - Control for Last Week Spending

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.022***	0.051***	0.007
	(0.008)	(0.008)	(0.008)
Close Loss	0.004	0.008	0.006
	(0.007)	(0.007)	(0.008)
Upset Win	-0.017	-0.042**	-0.010
	(0.017)	(0.017)	(0.019)
Predicted Win	-0.010*	-0.020***	0.013**
	(0.005)	(0.006)	(0.006)
Predicted Close	0.006	-0.002	0.002
	(0.006)	(0.006)	(0.007)
Predicted Loss	0.025***	0.023***	0.025**
	(0.009)	(0.009)	(0.010)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj.R^2	0.137	0.152	0.209
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last week and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A9: Emotional Shocks and Household Total Spending - Clustering at HH and season level

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
upset_loss	0.023***	0.052***	0.007
	(0.008)	(0.008)	(0.009)
close_loss	0.005	0.008	0.005
	(0.007)	(0.007)	(0.008)
upset_win	-0.013	-0.041**	-0.011
	(0.017)	(0.018)	(0.019)
Predicted Win	-0.010*	-0.021***	0.012**
	(0.006)	(0.006)	(0.006)
Predicted Close	0.005	-0.002	0.003
	(0.006)	(0.006)	(0.007)
Predicted Loss	0.026***	0.024***	0.025**
	(0.009)	(0.009)	(0.010)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.129	0.152	0.206
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household and game season level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A10: Emotional Shocks and Household Total Spending - Exclude HHs with over a month of zero spending

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.023***	0.053***	0.009
	(0.008)	(0.009)	(0.009)
Close Loss	0.009	0.011	0.011
	(0.007)	(0.008)	(0.008)
Upset Win	-0.001	-0.046**	-0.012
	(0.018)	(0.019)	(0.020)
Predicted Win	-0.011**	-0.022***	0.009
	(0.006)	(0.006)	(0.006)
Predicted Close	0.002	-0.004	-0.003
	(0.006)	(0.007)	(0.007)
Predicted Loss	0.021**	0.029***	0.025**
	(0.009)	(0.010)	(0.010)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.137	0.151	0.186
Observations	1,064,209	1,064,209	1,064,209

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household and game season level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A11: Emotional Shocks and Household Total Spending - Different Methods for Last Week's Shopping Data

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.025***	0.035***	-0.004
	(0.007)	(0.007)	(0.008)
Close Loss	0.001	0.000	0.001
	(0.006)	(0.007)	(0.007)
Upset Win	-0.010	-0.043***	0.024
	(0.015)	(0.016)	(0.017)
Predicted Win	-0.008	-0.021***	0.012**
	(0.005)	(0.005)	(0.006)
Predicted Close	0.002	0.000	-0.000
	(0.006)	(0.006)	(0.006)
Predicted Loss	0.021***	0.028***	-0.001
	(0.008)	(0.008)	(0.009)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.138	0.152	0.186
Observations	1,345,320	1,345,320	1,345,320

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table A12: Emotional Shocks and Household Total Spending - Team Fans Defined at the County Level

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.032**	0.070***	0.012
	(0.013)	(0.013)	(0.014)
Close Loss	0.013	0.010	-0.003
	(0.011)	(0.011)	(0.012)
Upset Win	0.004	-0.078***	0.016
	(0.025)	(0.026)	(0.029)
Predicted Win	-0.016*	-0.017*	0.004
	(0.009)	(0.009)	(0.010)
Predicted Close	-0.001	0.008	0.023**
	(0.009)	(0.009)	(0.010)
Predicted Loss	0.033**	0.027**	0.022
	(0.013)	(0.014)	(0.015)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.132	0.154	0.208
Observations	474,310	474,310	474,310

Notes: The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table A13: Emotional Shocks and Household Total Spending - Falsification Test

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Losss-following month	-0.005	0.018*	0.015
	(0.009)	(0.009)	(0.010)
Close Losss-following month	-0.003	0.001	0.009
	(0.006)	(0.006)	(0.007)
Upset Wins-following month	0.015	0.016	-0.013
	(0.010)	(0.010)	(0.011)
Predicted Win	-0.011	-0.019	-0.004
	(0.012)	(0.013)	(0.014)
Predicted Close	-0.014	-0.007	-0.011
	(0.012)	(0.012)	(0.013)
Predicted Loss	-0.016	-0.009	0.002
	(0.012)	(0.013)	(0.014)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.203	0.224	0.206
Observations	1,007,709	1,007,709	1,007,709

Notes: In this table, we investigated whether households' shopping decisions in a given week are impacted by the game results of the following month as a falsification test. The sample is restricted to expenditures following Sunday home and division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Appendix B Results for ALL Games and Home Games

Table B1: Summary Statistics of NFL Games, 2004-2019

	Number of Games	Fraction on Category or Subcategory (%)
Panel A. Sunday Regular Season NFL Games		
All Games	6,072	
Games Starting Before 1 PM	3,877	63.85
Games Starting Before 7 PM	5,610	92.39
Home Games	3,027	
Games Starting Before 1 PM	1,918	63.36
Games Starting Before 7 PM	2,795	92.34
Panel B. Sunday Regular Season NFL Games - All Games		
Outcome		
Loss	3,035	49.98
Win	3,037	50.02
Predicted and Actual Outcomes		
Predicted Win: Point Spread $\leq$ -4	1,650	27.17
Actual Win	1,222	74.06
Actual Loss (Upset Loss)	428	25.94
Predicted Close: -4 < Point Spread < 4	2,773	45.67
Actual Win	1,386	49.98
Actual Loss (Close Loss)	1,387	50.02
Predicted Loss: Point Spread $\geq$ 4	1,649	27.16
Actual Win (Upset Win)	429	26.02
Actual Loss	1,220	73.98
Salience Games		
(b) Sacks $\geq$ 5, Turnovers $\geq$ 5, or Penalties > 100 yds	3,212	52.90
Panel C. Sunday Regular Season NFL Games - Home Games		
Outcome		
Loss	1,302	43.01
Win	1,725	56.99
Predicted and Actual Outcomes		
Predicted Win: Point Spread $\leq$ -4	1,199	39.61
Actual Win	896	74.73
Actual Loss (Upset Loss)	303	25.27
Predicted Close: -4 < Point Spread < 4	1,386	45.79
Actual Win	709	51.15
Actual Loss (Close Loss)	677	48.85
Predicted Loss: Point Spread $\geq$ 4	442	14.60
Actual Win (Upset Win)	120	27.15
Actual Loss	322	72.85
Salience Games		
(b) Sacks $\geq$ 5, Turnovers $\geq$ 5, or Penalties > 100 yds	1,602	52.92

Notes: (1) This table presents summary statistics for NFL football games during the first 16 weeks of each season from 2004 to 2019. For each NFL game where both teams are included in the analysis, the game is recorded twice: once from the perspective of Team A and once from the perspective of Team B. (2) We exclude the New York Jets, New York Giants, Los Angeles Chargers, and Los Angeles Rams due to their shared home stadium locations. (3) Between 2004 and 2019, 18 regular-season NFL games played on Sundays ended in a tie. These ties were counted as a half-win and half-loss in league standings. We exclude these games in our analysis.

Table B2: Summary Statistics of Shopping Behaviors, 2004-2019

	All Game Week		Home Game Week		Bye Week	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Total Expenditure						
Sunday	17.12	48.22	17.09	48.09	17.39	47.94
Monday to Tuesday	20.14	43.59	20.11	43.47	19.90	43.07
Monday to Saturday	67.70	85.67	67.60	85.52	67.59	84.83
Food Expenditure						
Sunday	11.23	32.85	11.23	32.84	11.53	33.02
Monday to Tuesday	12.62	28.81	12.62	28.78	12.62	28.81
Monday to Saturday	42.41	54.48	42.36	54.39	42.88	54.75
Alcohol Expenditure						
Sunday	0.46	5.01	0.46	5.04	0.44	5.03
Monday to Tuesday	0.67	5.85	0.67	5.84	0.61	5.41
Monday to Saturday	2.49	12.05	2.49	11.94	2.31	11.54
Non-Food Expenditure						
Sunday	5.42	21.44	5.40	21.29	5.43	20.69
Monday to Tuesday	6.85	22.25	6.83	22.13	6.67	21.52
Monday to Saturday	22.80	44.04	22.75	44.08	22.40	42.69
Shopping Trips[a]						
Sunday	0.28	0.45	0.28	0.45	0.29	0.45
Monday to Tuesday	0.47	0.64	0.47	0.64	0.46	0.64
Monday to Saturday	1.46	1.34	1.46	1.34	1.46	1.33
Sample Size	4,955,746		2,459,866		388,839	

Notes: The summary statistics are unweighted for broader representativeness, as the final sample cannot be extrapolated to national, regional, or scantrack market levels where projection factors were applied in the HomeScan data. This table primarily presents the means and standard deviations of household expenditures and shopping trips following Sunday all games, home games, or bye weeks during regular seasons. We deflate expenditures to 2012 dollars using the consumer price index for urban consumers for all items. [a]Shopping trips are defined by the frequency of visits made to purchase groceries. Each visit is counted as one trip, regardless of whether it involves stops at multiple stores.

Table B3: The Effects of Emotional Shocks on Household Total Spendings

Dependent Var: Ln (total spending on...)	All Games			Home Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
(a) Upset Loss	0.005	0.010***	-0.001	0.009**	0.016***	0.003
	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Close Loss	0.004*	0.003	0.002	0.011***	0.010***	0.007*
	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
(b) Upset Win	-0.002	-0.014***	-0.012***	-0.007	-0.020**	-0.025***
	(0.004)	(0.004)	(0.004)	(0.008)	(0.008)	(0.009)
Predicted Win	-0.010***	-0.007*	0.007*	-0.010**	-0.010**	0.008**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Close	-0.006*	0.002	0.002	-0.010***	0.003	0.002
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Loss	-0.003	0.008**	0.008*	0.007	0.016***	0.015**
	(0.004)	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
p-value for row (a) = -row (b)	0.605	0.490	0.028	0.844	0.549	0.032
Adj. R^2	0.130	0.158	0.207	0.130	0.157	0.206
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's childbearing, marital status, race, and internet access, as well as household's age, education years, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** p<0.01, ** p<0.05, * p<0.1.

Table B4: Emotional Shocks and Household Shopping Trips

Dependent Var: shopping trips	All Games			Home Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.002	0.006**	-0.000	0.005	0.008***	-0.001
	(0.003)	(0.002)	(0.001)	(0.003)	(0.003)	(0.002)
Close Loss	0.003	0.002	0.001	0.009***	0.005**	0.002
	(0.002)	(0.002)	(0.001)	(0.003)	(0.002)	(0.002)
Upset Win	-0.002	-0.008***	-0.004***	-0.008	-0.014***	-0.007**
	(0.003)	(0.003)	(0.002)	(0.006)	(0.005)	(0.003)
Predicted Win	-0.008***	-0.004*	0.001	-0.007**	-0.005*	0.002
	(0.003)	(0.002)	(0.001)	(0.003)	(0.003)	(0.002)
Predicted Close	-0.006**	0.003	0.001	-0.009***	0.004	0.002
	(0.003)	(0.002)	(0.001)	(0.003)	(0.003)	(0.002)
Predicted Loss	-0.003	0.006**	0.004**	0.003	0.009**	0.006***
	(0.003)	(0.002)	(0.002)	(0.004)	(0.004)	(0.002)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,268,701	5,273,141	5,338,184	2,759,147	2,774,102	2,839,308

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the first 16 weeks of the regular NFL football season from 2004 to 2019. Given the incident-based nature of trip data, we use the Poisson model (`PPMLHDFE` command in STATA 18) for the number of trips reported by households. Some observations are excluded because they are either singletons or separated due to fixed effects. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (e.g., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B5: Emotional Shocks and Household Spendings on Different Categories

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Panel A. Food Spending						
Upset Loss	0.003	0.007**	-0.001	0.007*	0.014***	0.004
	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.005)
Close Loss	0.004*	0.003	0.002	0.009***	0.011***	0.005
	(0.002)	(0.002)	(0.003)	(0.003)	(0.004)	(0.004)
Upset Win	-0.003	-0.014***	-0.016***	-0.009	-0.020***	-0.029***
	(0.004)	(0.004)	(0.004)	(0.007)	(0.008)	(0.009)
Predicted Win	-0.008**	-0.007*	0.007*	-0.007**	-0.009***	0.008**
	(0.003)	(0.003)	(0.004)	(0.003)	(0.004)	(0.004)
Predicted Close	-0.006*	-0.001	0.003	-0.009**	-0.001	0.003
	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Loss	-0.003	0.006*	0.009**	0.006	0.014***	0.018***
	(0.003)	(0.004)	(0.004)	(0.005)	(0.005)	(0.006)
Panel B. Alcohol Spending						
Upset Loss	0.001	0.000	-0.001	0.002	0.003**	0.001
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)
Close Loss	0.001*	0.001	0.000	0.001	0.001	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)
Upset Win	0.001	-0.002**	-0.003	-0.003	-0.003	-0.010**
	(0.001)	(0.001)	(0.002)	(0.002)	(0.003)	(0.004)
Predicted Win	-0.001	-0.001	-0.001	-0.000	-0.003**	-0.003*
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)
Predicted Close	-0.001	-0.001	-0.000	-0.001	-0.000	0.002
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)
Predicted Loss	-0.001	0.000	-0.001	0.002	0.003	0.005**
	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)	(0.003)
Panel C. Non-food Spending						
Upset Loss	0.004	0.005*	-0.002	0.007**	0.005	-0.000
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)
Close Loss	0.002	0.001	0.003	0.007**	0.004	0.007*
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.004)
Upset Win	0.001	-0.005*	-0.002	0.004	-0.010	-0.017**
	(0.004)	(0.003)	(0.004)	(0.007)	(0.007)	(0.008)
Predicted Win	-0.006**	-0.002	0.001	-0.005	-0.005*	0.001
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)
Predicted Close	-0.002	0.001	0.001	-0.002	0.002	0.003
	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Predicted Loss	0.003	0.004*	0.008*	0.007	0.009**	0.013**
	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.005)
Observations	5,344,585	5,344,585	5,344,585	5,344,585	5,344,585	5,344,585

Notes: Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. Food includes dry grocery, frozen foods, dairy, deli, fresh produce, and packaged meat departments. Non-food includes health & beauty care, non-food grocery, and general merchandise departments. These categorizations are from the consumer panel data collected by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B6: Emotional Shocks and Household Shopping on Different Types of Food

Dependent Var: Ln (spending on...)	All Games			Home Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Panel A. Perishable Food						
Upset Loss	0.002	0.003	-0.000	0.006**	0.010***	0.003
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.004)
Close Loss	0.002	0.002	0.002	0.007***	0.010***	0.006*
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)
Upset Win	-0.002	-0.012***	-0.013***	-0.009*	-0.015***	-0.019***
	(0.003)	(0.003)	(0.003)	(0.005)	(0.005)	(0.007)
Predicted Win	-0.002	-0.003	0.006**	-0.001	-0.006**	0.007**
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Predicted Close	-0.001	-0.000	0.003	-0.004	-0.002	0.003
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Predicted Loss	0.002	0.004*	0.008***	0.008**	0.007**	0.010**
	(0.002)	(0.003)	(0.003)	(0.004)	(0.004)	(0.005)
Panel B. Non-perishable Food						
Upset Loss	0.003	0.007**	0.000	0.006*	0.013***	0.005
	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Close Loss	0.004*	0.003	0.002	0.007**	0.009***	0.004
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.004)
Upset Win	-0.004	-0.013***	-0.014***	-0.010	-0.019***	-0.026***
	(0.003)	(0.003)	(0.004)	(0.007)	(0.007)	(0.008)
Predicted Win	-0.007**	-0.006*	0.005	-0.007**	-0.008**	0.007*
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)
Predicted Close	-0.005*	-0.002	0.001	-0.008**	-0.001	0.002
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
Predicted Loss	-0.002	0.005	0.007*	0.006	0.013***	0.016***
	(0.003)	(0.003)	(0.004)	(0.004)	(0.005)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. Perishable food includes dairy, deli, fresh produce, and packaged meat departments. Nonperishable food includes dry grocery and frozen foods departments. These categorizations are based on product hierarchy structured by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table B7: Shocks from Emotionally Salient Games

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Panel A. More Salient Games						
Upset Loss	0.005	0.017***	-0.001	0.007	0.030***	0.006
	(0.005)	(0.005)	(0.006)	(0.006)	(0.006)	(0.007)
Close Loss	0.001	0.007**	0.001	0.007	0.005	0.012**
	(0.003)	(0.004)	(0.004)	(0.005)	(0.005)	(0.006)
Upset Win	0.001	-0.011*	0.004	-0.023*	-0.039***	-0.012
	(0.006)	(0.006)	(0.006)	(0.012)	(0.013)	(0.013)
Predicted Win	-0.012***	-0.010**	0.008*	-0.010**	-0.016***	0.005
	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Predicted Close	-0.002	-0.005	-0.000	-0.005	-0.005	-0.003
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)
Predicted Loss	0.000	0.007	0.006	0.016**	0.028***	0.023***
	(0.005)	(0.005)	(0.005)	(0.007)	(0.008)	(0.008)
Observations	2,955,412	2,955,412	2,955,412	1,674,715	1,674,715	1,674,715
Panel B. Less Salient Games						
Upset Loss	0.004	0.004	-0.000	0.010	0.004	-0.001
	(0.005)	(0.005)	(0.006)	(0.006)	(0.007)	(0.007)
Close Loss	0.006*	-0.002	0.003	0.016***	0.015**	-0.001
	(0.004)	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)
Upset Win	-0.001	-0.020***	-0.025***	0.013	-0.018	-0.025*
	(0.006)	(0.006)	(0.006)	(0.011)	(0.012)	(0.013)
Predicted Win	-0.005	-0.005	0.009*	-0.008	-0.005	0.015***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)
Predicted Close	-0.008*	0.007	0.009*	-0.012**	0.006	0.014**
	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)
Predicted Loss	-0.003	0.010**	0.012**	0.002	0.006	0.007
	(0.005)	(0.005)	(0.005)	(0.007)	(0.008)	(0.008)
Observations	2,768,599	2,768,599	2,768,599	1,549,115	1,549,115	1,549,115
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the impact of three potentially frustrating events: five or more sacks, five or more turnovers, or 100 or more penalty yards. The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B8: Emotional Shocks and Household Total Spending - Different Cutoffs

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.004	0.012***	-0.001	0.010***	0.020***	0.004
	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Close Loss	0.000	-0.007*	0.006	0.012**	0.012**	0.014**
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.006)
Upset Win	-0.006**	-0.011***	-0.006*	-0.011*	-0.006	-0.012*
	(0.003)	(0.003)	(0.003)	(0.005)	(0.006)	(0.006)
Predicted Win	-0.009***	-0.007*	0.007*	-0.009**	-0.009**	0.008*
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Close	-0.005	0.010**	-0.005	-0.011**	0.007	-0.004
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)
Predicted Loss	-0.001	0.008**	0.008*	0.004	0.008	0.012**
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.130	0.158	0.207	0.130	0.157	0.206
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -3 or less, *Predicted Close* indicates a point spread between -3 and 3 (exclusive), and *Predicted Loss* indicates a spread of +3 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B9: Emotional Shocks and Household Total Spending - Different Fixed Effects

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.012***	0.012***	0.004	0.012**	0.011**	0.006
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)
Close Loss	0.002	-0.000	0.000	0.006	0.006	0.006
	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Upset Win	0.002	-0.004	-0.008*	0.010	-0.014	-0.016*
	(0.004)	(0.004)	(0.005)	(0.009)	(0.009)	(0.010)
Predicted Win	-0.017***	-0.003	0.006	-0.012***	-0.002	0.009**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Predicted Close	-0.007**	0.005	0.002	-0.009**	0.004	0.002
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Predicted Loss	-0.004	0.007*	0.006	0.001	0.016**	0.014**
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season - week FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.131	0.160	0.207	0.131	0.159	0.207
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B10: Emotional Shocks and Household Total Spending - Control for Last Week Spending

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.004	0.010***	-0.001	0.008*	0.016***	0.003
	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Close Loss	0.004	0.002	0.003	0.011***	0.010***	0.008*
	(0.002)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Upset Win	-0.003	-0.014***	-0.012***	-0.010	-0.021**	-0.023**
	(0.004)	(0.004)	(0.004)	(0.008)	(0.008)	(0.009)
Predicted Win	-0.010***	-0.007*	0.007*	-0.010**	-0.010**	0.008**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Close	-0.006*	0.002	0.002	-0.010***	0.002	0.002
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Loss	-0.002	0.008**	0.007*	0.007	0.015***	0.014**
	(0.004)	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.139	0.157	0.209	0.138	0.156	0.209
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B11: Emotional Shocks and Household Total Spending - Clustering at HH and season level

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.005	0.010***	-0.001	0.009**	0.016***	0.003
	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Close Loss	0.004*	0.003	0.002	0.011***	0.010***	0.007*
	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Upset Win	-0.002	-0.014***	-0.012***	-0.007	-0.022***	-0.025***
	(0.004)	(0.004)	(0.004)	(0.008)	(0.008)	(0.009)
Predicted Win	-0.010***	-0.007*	0.007*	-0.010**	-0.010**	0.008**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Close	-0.006*	0.002	0.002	-0.010***	0.003	0.002
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Loss	-0.003	0.008**	0.008*	0.007	0.016***	0.015**
	(0.004)	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.130	0.158	0.207	0.130	0.157	0.206
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household and game season level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B12: Emotional Shocks and Household Total Spending - Exclude HHs with over a month of zero spending

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.005	0.011***	0.001	0.010**	0.017***	0.005
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.005)
Close Loss	0.006**	0.003	0.002	0.013***	0.012***	0.007*
	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Upset Win	-0.004	-0.016***	-0.013***	-0.006	-0.026***	-0.028***
	(0.004)	(0.004)	(0.005)	(0.009)	(0.009)	(0.009)
Predicted Win	-0.013***	-0.008**	0.003	-0.011***	-0.011**	0.005
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Close	-0.008**	0.002	0.000	-0.013***	0.002	0.001
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Predicted Loss	-0.003	0.012***	0.010**	0.005	0.020***	0.018***
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.137	0.155	0.186	0.138	0.154	0.186
Observations	4,834,408	4,834,408	4,834,408	2,577,009	2,577,009	2,577,009

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B13: Emotional Shocks and Household Total Spending - Different Methods for Last Week's Shopping Data

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.009**	0.008**	-0.001	0.011***	0.014***	0.000
	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Close Loss	0.003	0.000	0.000	0.011***	0.007*	0.004
	(0.002)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Upset Win	0.002	-0.018***	-0.011***	-0.008	-0.023***	-0.009
	(0.004)	(0.004)	(0.004)	(0.008)	(0.008)	(0.009)
Predicted Win	-0.012***	-0.006	0.006	-0.009**	-0.010***	0.009**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Close	-0.008**	0.000	0.004	-0.012***	0.002	0.003
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Predicted Loss	-0.004	0.007*	0.007	0.007	0.019***	0.007
	(0.004)	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.139	0.156	0.186	0.140	0.155	0.186
Observations	5,746,053	5,746,053	5,746,053	3,049,896	3,049,896	3,049,896

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B14: Emotional Shocks and Household Total Spending - Team Fans Defined at the County Level

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.012**	0.012**	0.005	0.015**	0.011	0.009
	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)	(0.008)
Close Loss	-0.001	-0.002	0.000	0.010*	0.007	0.000
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.007)
Upset Win	-0.004	-0.024***	-0.011	-0.017	-0.031**	-0.021
	(0.006)	(0.007)	(0.007)	(0.013)	(0.013)	(0.014)
Predicted Win	-0.021***	-0.005	0.008	-0.020***	-0.008	0.005
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)
Predicted Close	-0.010*	0.005	0.008	-0.017***	0.008	0.013*
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)
Predicted Loss	-0.007	0.010	0.008	0.010	0.024***	0.015
	(0.006)	(0.006)	(0.007)	(0.009)	(0.009)	(0.010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.135	0.160	0.209	0.134	0.159	0.209
Observations	2,123,409	2,123,409	2,123,409	1,136,867	1,136,867	1,136,867

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B15: Emotional Shocks and Household Total Spending - Falsification Test

Dependent Var:	All Games			Home Games		
Ln(spending on...)	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss-following month	-0.000	-0.004	-0.009*	-0.003	0.006	-0.002
	(0.004)	(0.004)	(0.005)	(0.005)	(0.006)	(0.006)
Close Loss-following month	-0.001	0.000	0.003	0.002	0.001	0.004
	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Upset Win-following month	0.000	-0.001	0.001	0.003	-0.001	-0.007
	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)	(0.007)
Predicted Win-following month	0.010**	-0.006	0.000	0.021***	-0.008	0.001
	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)	(0.007)
Predict Close-following month	0.005	-0.004	-0.005	0.013**	-0.007	-0.005
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.006)
Predict Loss-following month	0.012***	-0.004	-0.002	0.020***	-0.003	0.002
	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)	(0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R^2	0.133	0.161	0.205	0.133	0.160	0.205
Observations	4,419,981	4,419,981	4,419,981	2,397,646	2,397,646	2,397,646

Notes: In this table, we investigated whether households' shopping decisions in a given week are impacted by the game results of the following month as a falsification test. The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.